Applied Soft Computing Journal 162 (2024) 111770

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

A dual-population coevolutionary algorithm for balancing convergence and diversity in the decision space in multimodal multi-objective optimization

Zhipan Li [a], Huigui Rong [a,*], Shengxiang Yang [b], Xu Yang [c], Yupeng Huang [a]

[a] The College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China
[b] School of Computer Science and Informatics, De Montfort University, Leicester LE1 9BH, UK
[c] College of Computer, National University of Defense Technology, Changsha 410073, China

ARTICLE INFO

Keywords:
Multimodal multi-objective
Evolutionary algorithm
Global convergence-first
Dual-population
Coevolutionary

ABSTRACT

Many multimodal multi-objective evolutionary algorithms (MMEAs) are effective in solving multimodal multi-objective problems (MMOPs), which have multiple equivalent Pareto optimal sets (PSs) mapping to the same Pareto optimal front (PF). Due to the existence of the global convergence-first mechanism, these MMEAs will remove the solutions that can improve the diversity of the decision space but have poor convergence and even lead to the loss of PS when encountering MMOPs with an imbalance between convergence and diversity in the decision space (MMOP-ICD) or an MMOP with a local PS (MMOPL). We propose a new dual-population coevolutionary algorithm to address these issues. The auxiliary population helps the main population locate areas where equivalent PSs may exist, and the main population focuses on balancing convergence and diversity in the decision space. When updating the auxiliary population, a strength local convergence quality (SLCQ) is used to explore the distribution of the equivalent PSs. When updating the main population, the new niche-based truncation strategy first deletes the solutions that contribute less to convergence. Then, a distance-based subset selection method balances the diversity between the decision and objective spaces. The comparison results show the overall performance of the proposed algorithm is significantly better than other state-of-the-art algorithms.

1. Introduction

Multi-objective optimization problems (MOPs) widely exist in practical applications such as in workshop scheduling [1], supply chain management [2] and multi-objective knapsack optimization [3]. MOPs have multiple conflicting objectives that must be optimized simultaneously. Without loss of generality, a minimization MOP can be formulated as [4]:

$$\begin{cases} \text{Minimize } F(X) = (f_1(X), f_2(X), \ldots, f_m(X)), \\ \text{s.t.} \quad X = (x_1, x_2, \ldots, x_n) \in \Omega, \end{cases} \tag{1}$$

where Ω denotes the decision space; X is an n-dimensional decision variable vector and $F(X)$ consists of m conflicting objective functions. In MOPs, for any two feasible solutions, $\vec{x}$ and $\vec{y}$, if $f_i(\vec{x}) \leq f_i(\vec{y})$ for each i ($i \in 1, \ldots, m$) and there exists at least one j ($j \in 1, \ldots, m$) so that $f_j(\vec{x}) < f_j(\vec{y})$, then $\vec{x}$ is considered to dominate $\vec{y}$. A solution is a Pareto optimal solution if no solutions dominate it. The set of Pareto optimal solutions is called the Pareto optimal set (PS), and the image of the PS in the objective space is called the Pareto optimal front (PF) [5].

In some MOPs, multiple equivalent decision vectors may map the same objective vector. This kind of MOP is the multimodal multi-objective optimization problem (MMOP) [6]. Through Fig. 1, we introduce MMOP and some of its special forms. In the standard form of MMOP in the initial phase, solutions A_1 and A_2 are far apart in the decision space but map to the same objective vector. As more practical application scenarios are found to have the characteristics of the MMOP, some MMOPs may have an imbalance between convergence and diversity in the decision space (MMOP-ICD) [7], and different PSs of one MMOP-ICD have different search difficulties. As shown in Fig. 1, the distance between A_1 and B_1 is θ. The distance between A_2 and B_2 is also θ, but B_1 dominates B_2 in the objective space, so A_1 is relatively easy to search, and A_2 is relatively difficult to search. There have been MMOPs with a local PS (MMOPL) [8]. It can be seen from Fig. 1 that although the solution A_3 is dominated by the solutions A_1 and A_2, A_3 is close to A_1 and A_2 in the objective space. Since A_3 has similar objective values to A_1 and A_2, A_1, A_2, and A_3 are equivalent solutions within the range acceptable to the decision-maker. Multimodal multi-objective

* Corresponding author.
E-mail address: ronghg@hnu.edu.cn (H. Rong).

https://doi.org/10.1016/j.asoc.2024.111770
Received 20 September 2023; Received in revised form 8 April 2024; Accepted 13 May 2024
Available online 28 May 2024
1568-4946/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

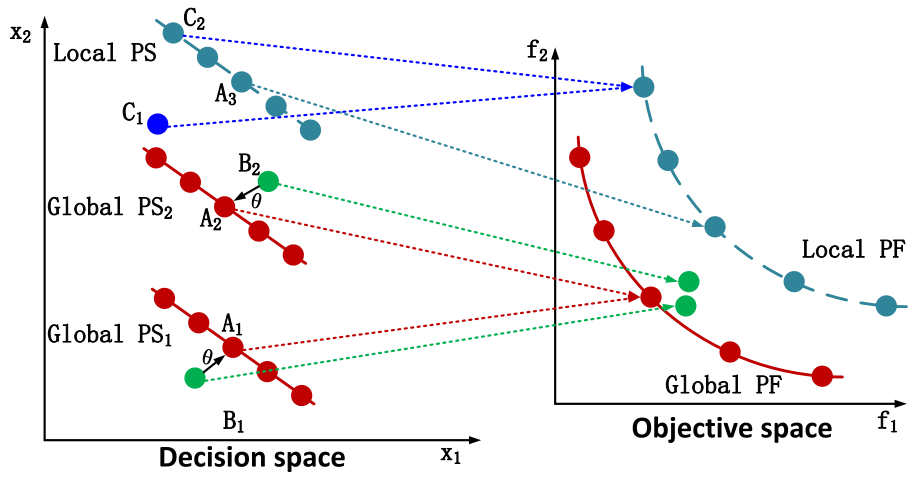


Fig. 1. Illustration of the two-objective MMOP.

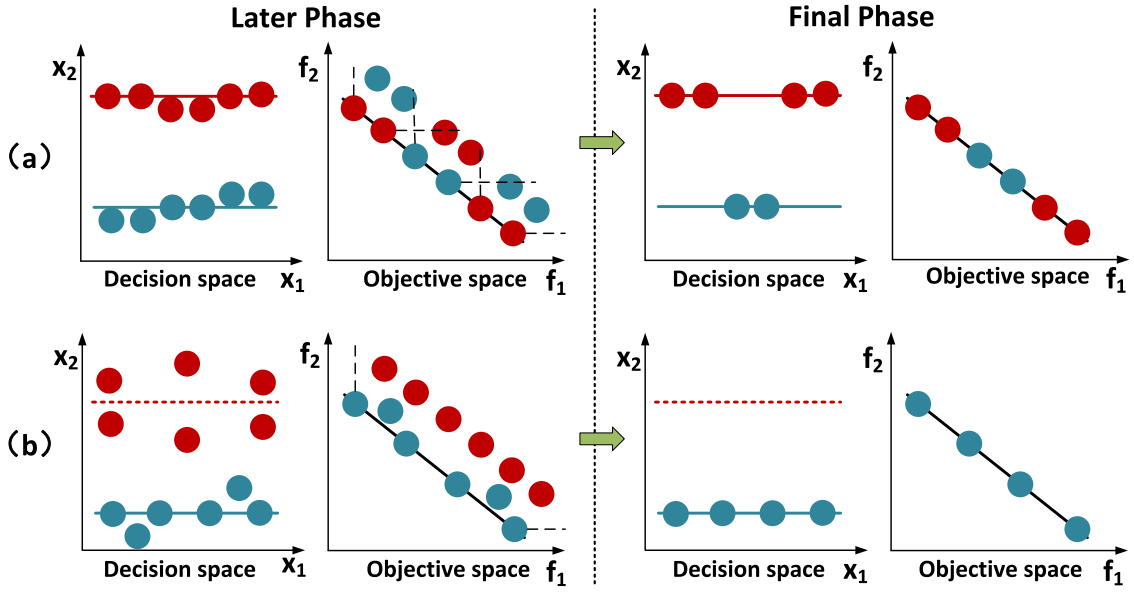


Fig. 2. The effect of global convergence-first mechanism. Figure (a) depicts the variation in diversity in the decision space. Figure (b) depicts the phenomenon of PS loss.

optimization (MMO) aims to find these equivalent PSs so that decision-makers can choose appropriate PSs according to the characteristics of practical application problems or the preferences of decision-makers.

Over the past two decades, many multi-objective evolutionary algorithms (MOEAs) have been proposed. Typical representative algorithms include the nondominated sorting genetic algorithm II [9], the strength Pareto evolutionary algorithm [10], and the MOEA based on decomposition [11]. When MMOPs first appeared, these MOEAs were found to be ineffective in solving MMOPs. Since they only pay attention to the convergence-first mechanism of the objective space, the solutions that improve the diversity of the decision space but have poor convergence are deleted. In the later phase of Fig. 2(a), because the solutions with slightly poorer convergence are removed, the diversity of the two PSs obtained in the final stage becomes very poor.

Aiming at the shortcomings of MOEAs in solving MMOPs, scholars have proposed many multimodal multi-objective evolutionary algorithms (MMEAs) to solve MMOPs. For example, the MMEA with dynamic neighbors [12], the MMEA with double-niched [13], the multi-objective particle swarm optimization algorithm using ring topology and special crowding distance (MO_Ring_PSO_SCD) [14], the MMEA using two-archive and recombination strategies (TriMOEA-TA&R) [15]. They add the diversity maintenance mechanism to the decision space to balance the diversity between the objective and decision spaces, thereby alleviating the imbalance between diversity and convergence in the decision space. When retaining solutions, these MMEAs keep solutions close in the objective space but far away in the decision space. Although the deleted solutions in Fig. 2(a) have slightly poorer convergence, these MMEAs retain them in order to preserve diversity in the decision space. Since the global convergence-first mechanism still exists in these MMEAs, these MMEAs have poor diversity in the decision space when solving MMOPs.

With the emergence of MMOP-ICD and MMOPL, preserving the imbalanced PS and the local PS brings significant challenges to the MMEAs. In Fig. 2(b), the red-dotted line can represent a local PS or an imbalanced PS that is more difficult to search. During the search process, the convergence quality of the solutions around the red-dotted PS is worse than those near the blue PS. The MMEAs with a global convergence-first mechanism trim the solutions near the red-dotted PS. As a result, the algorithm cannot search for the local PS and the imbalanced PS.

For the MMOP-ICD, popular algorithms include the MMEA with convergence-penalized density strategy (CPDEA) [7], the two-archive MMEA with decomposition and fitness allocation [16], and weighted indicator-based MMEA (MMEA-WI) [17]. These MMEAs relax the global

convergence-first mechanism, and this convergence-relaxed mechanism allows these MMEAs to retain solutions near the imbalanced PS with more incredible search difficulty. Therefore, these MMEAs based on a convergence-relaxed mechanism can search for the imbalanced PS. However, since these MMEAs are for searching the global PS, the local PS will be lost when solving the MMOPL.

In order to search for a local PS, some scholars have enhanced the degree of relaxation of the convergence-first mechanism. Examples include the novel MMEA with dual clustering in the decision and objective spaces (MMOEA/DC) [18], the clustering-based MMEA [19], and the multi-population MMEA based on grid search [20]. They narrow the comparison of the Pareto dominance relationship between solutions within a cluster or a niche rather than globally, so the search for a local PS will not be affected by the global PS. However, as the degree of relaxation of the convergence-relaxed mechanism increases, when the gap in search difficulty between PSs of the MMOP-ICD becomes larger, these MMEAs will still easily lose the imbalanced PS with greater search difficulty. Furthermore, in Fig. 1, although solution C_1 and solution C_2 map to the same location of the Local PF, C_1 is not a true local Pareto optimum solution because there is a global PS near C_1. Thus C_1 is a pseudo-local Pareto optimal solution. However, the existing algorithms retain pseudo-local Pareto optimal solutions during evolution.

Based on the shortcomings of the existing work, we propose a dual-population coevolutionary algorithm for balancing convergence and diversity in the decision space, termed CDP-BCD. When updating the auxiliary population, CDP-BCD uses a strength local convergence quality evaluation method to localize the positions of equivalent PSs, especially imbalanced and local PSs. The better the SLCQ of the solution, the closer the solution is to the true PS. When updating the main population, solutions that do not contribute much to the convergence of the population are deleted through a niche-based truncation method. Then, a distance-based subset selection scheme is used to balance the diversity between the decision and objective spaces. The main contributions of this study are as follows:

(1) An imbalance between diversity and convergence in the decision space due to too strong interactions between populations in traditional strong and weak interaction coevolutionary mechanisms, we propose a new dual-population weak interaction coevolutionary mechanism (DPCE). DPCE weakens the intensity of interaction between populations in traditional weak interaction coevolutionary mechanisms. In the early stage of the algorithm, the auxiliary population is used to explore the locations of the global PS and the local PS. In the later stage of the algorithm, the main population and the auxiliary population cooperate to develop locations where the global PS and the local PS exist.
(2) Due to the insufficient selection pressure on convergence caused by existing local convergence quality evaluation methods, we propose an SLCQ evaluation method. SLCQ combines the convergence quality of a solution within its neighborhood and the distance to the neighboring solutions that dominate it. SLCQ gives higher priority in environmental selection to solutions that have slightly poorer global convergence quality but contribute significantly to the diversity of the decision space.
(3) We propose a niche-based truncation strategy (NBT). Since SLCQ represents the convergence of the solution in a local area in the decision space, there will be many similar solutions with the same SLCQ value in this local area. Computational resources are wasted when a local area is exploited and explored by multiple similar solutions simultaneously. Therefore, NBT uses a niche to filter out and delete the solutions that do not contribute much to the convergence.
(4) We propose a new distance-based subset selection (DSS) scheme. Since the traditional methods for balancing the diversity between the decision and objective spaces require setting parameters, the algorithm's robustness will worsen. DSS adapts according to distance while balancing the diversity between decision and objective spaces without requiring additional parameters, which helps the algorithm adapt to different scenarios.

The remainder of this paper is organized as follows. In Section 2, the theoretical knowledge used to propose the CDP-BCD is introduced. The details of CDP-BCD are presented in Section 3. Experimental design and results are discussed in Section 4 and Section 5, respectively. Section 6 concludes this paper.

2. Related works and motivation

2.1. Related definitions of the MMOP

The MMOP has been widely studied, especially in recent years, scholars have conducted in-depth research on two special forms of MMOP including MMOPL and MMOP-ICD. The definitions of MMOPL and MMOP-ICD are as follows [6,7]:

Definition 1. An MMOP can be recognized as an MMOPL if it meets the following condition:

In the MMOP, two different solutions a and b are considered equivalent if $\|F(a) - F(b)\| \leq \theta$, where $\theta > 0$.

Definition 2. An MMOP can be defined as an MMOP-ICD if it meets one or both of the following two conditions:

(1) For the same point on PF, solutions close to one equivalent Pareto-optimal solution are more likely to be dominated by solutions close to another equivalent Pareto-optimal solution.
(2) For the same point on PF, the difficulty of searching for one equivalent Pareto optimal solution is higher than that of another equivalent Pareto optimal solution.

In the construction of the MMOP-ICD, the search difficulty between PSs is made different by assigning different difficulty functions to different PSs. When the difficulty function reaches the minimum value during evolution, the population converges to the true PS. The imbalanced distance minimization problem (IDMP) [7] belongs to the MMOP-ICD. The IDMP is to control the difficulty of the difficulty function to reach the minimum value to control the search difficulty of each PS (convergence difficulty). Fig. 3 shows the variation curves of different difficult functions with the decision variable x in some IDMPs. For the two difficult functions $g1$ and $g2$ of IDMPM2T1, since the slope of $g1$ is gentler than that of $g2$, the minimum value of $g1$ is easier to search, so the search difficulty of the PS bound to $g1$ is less, while the PS bound to g2 is difficult to search. For IDMPM3T1, since the search of the global minimum of $g1$ is affected by a small number of local minima, the global minimum of $g1$ will be easily searched, and the PS bound to $g1$ is also relatively easy to be searched. In contrast, the search for the global minimum of $g2$ is affected by a larger number of local minima, so the PS bound to $g2$ is more difficult to search.

2.2. Multimodal multi-objective evolutionary algorithm

In the MMO community, many scholars initially searched for multiple equivalent PSs by balancing the diversity between the decision and objective spaces. The MMEA with dynamic neighbors [12] uses the niche method to select some solutions that help increase the decision space's diversity when matching selections to search for multiple equivalent PSs. The double-niched MMEA [13] uses double-niche technology to balance diversity between the decision and objective spaces. The modified dominance relationship in the MO_Ring_PSO_SCD [14] enables the algorithm to search for multiple PSs and balances diversity between the decision and objective spaces. TriMOEA-TA&R [15] enables the algorithm to search for more PSs by classifying decision variables and exchanging independent convergence variables. The MMEA based on improved crowding distance [21] modifies the special

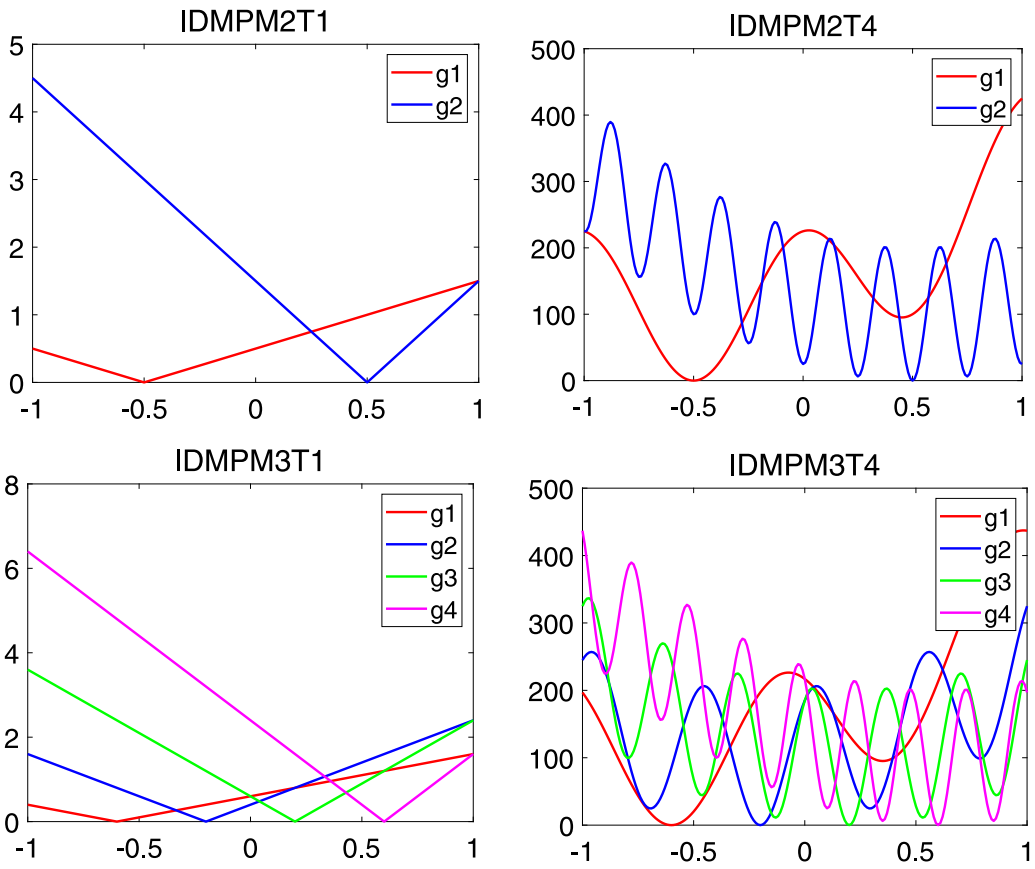


Fig. 3. Difficulty functions in some IDMPs. *g* represents the variation curve of the difficulty function with the decision variable x.

crowding distance in MO_Ring_PSO_SCD, so it has better diversity in the decision space.

With the emergence of MMOP-ICD, many algorithms that relax the convergence-first mechanism have been proposed (named convergence-weak-relaxed mechanism). CPDEA [7] and MMEA-WI [17] use the convergence quality indicator to punish the solution that converges too closely to the PS with less search difficulty so that the algorithm can search for the imbalanced PS with greater search difficulty. The MMEAs based on the modified Pareto dominance relationship [22,23] give the solutions around the imbalanced PS the opportunity to continue exploring the imbalanced PS by weakening the comparative strength of the Pareto dominance relationship between solutions. The MMEA based on independently evolving sub-problems [24], the two-archive algorithm with decomposition [16], and the MMEA with decomposition and subset selection [25] can search for imbalanced PS by finding solutions with good convergence and diversity in sub-problems. These algorithms all have good performance in solving MMOP-ICD.

To meet the decision-maker's needs for local PS, a new convergence-relaxed mechanism emerges, which mostly selects solutions with good convergence in the local range rather than solutions with good convergence in the global range (named convergence-strong-relaxed mechanism). The MMEA with hierarchy ranking method (HREA) [8] divides the global PS and local PS into different layers by the hierarchy ranking method and then prunes each layer independently to search for the local PS. The ring-hierarchy-based MMEA (RHEA) [26] uses a ring topology in the evolution process and only evolves the solutions of the adjacent nondominated layers each time, which can avoid the influence of the global PS on the search for the local PS. The clearing-based MMEA [27] uses clearing-based niche technology to remove poor-quality Pareto optimal solution sets to retain local PSs with good convergence quality. The two-stage MMEAs [28–30] allow the local PS to be retained during the evolution process through the exploration of the local PS in the first stage and the development of the global and local PS through niche or multi-population methods in the second stage. The grid-guided particle swarm optimizer [31] avoids the impact of the global PS on the search for the local PS through the independent evolution of grid sub-regions.

2.3. Multi-population coevolutionary mechanism

It is worth noting that there are currently many coevolutionary MOEAs, which can be divided into strong-interaction coevolutionary MOEAs and weak-interaction coevolutionary MOEAs according to the intensity of interaction [32]. One thing that strong interaction coevolution MOEAs have in common is that the interaction between the two populations is very strong [15,16,33]. As shown in Fig. 4(a), strong interaction coevolutionary MOEAs enable populations to cooperate in mating selection, offspring generation, and environmental selection. One thing that weak interaction coevolutionary MOEAs have in common is that the interaction strength between the two populations is very small [8,32,34]. As shown in Fig. 4(b), populations in weak interaction coevolutionary MOEAs evolve individually and only share all the offspring in each generation.

2.4. Motivation

Recently, many different types of MMOPs have been discovered from real-world engineering applications, including MMOP-NON-ICD [16], MMOP-ICD [7], and MMOPL [6]. Therefore, when designing the algorithm, it is necessary to ensure it can adapt to different types of MMOPs. To make the algorithm adaptable to different types of

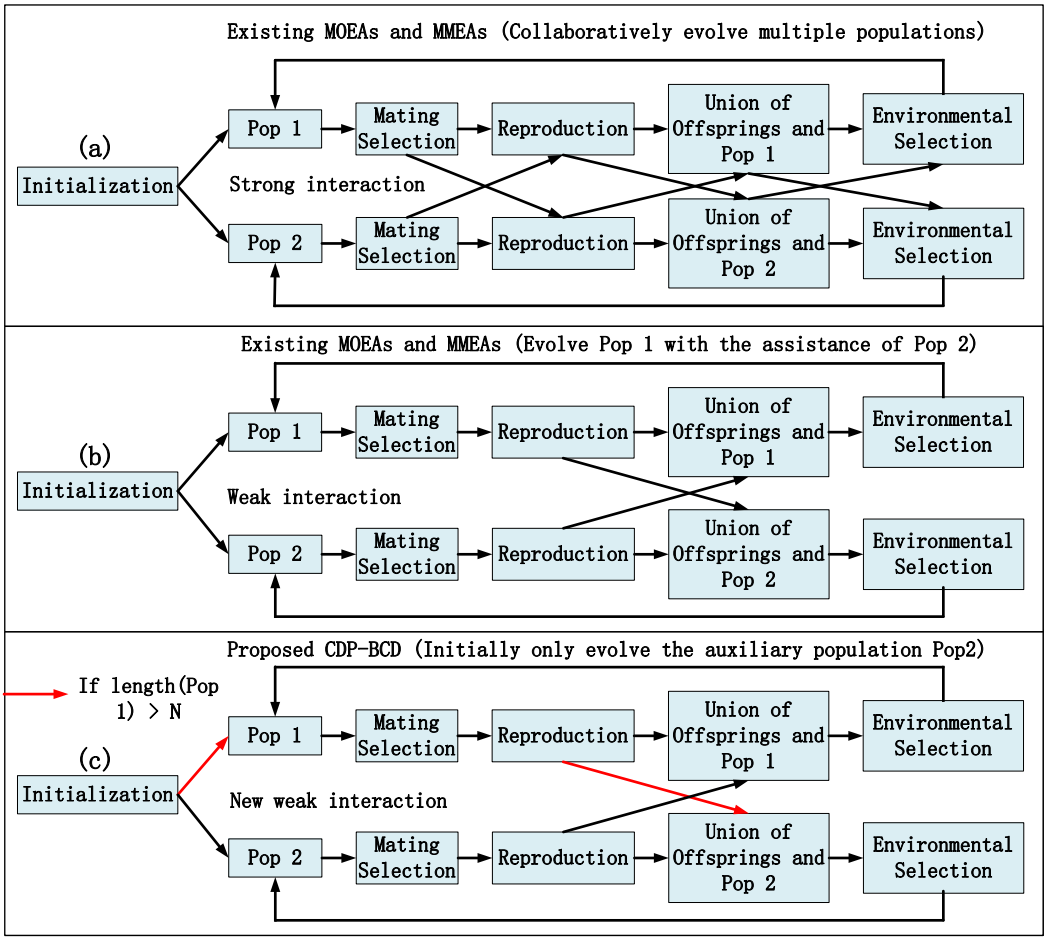


Fig. 4. Procedures of traditional strong interaction coevolutionary mechanism and the proposed weak interaction coevolutionary mechanism.

MMOPs, many scholars have designed more robust MMEAs by relaxing the convergence-first mechanism, such as modifying the dominance relationship or penalizing convergence. The initial typical excellent algorithms based on the convergence-weak-relaxed mechanism include CPDEA [7], DNPD [22], and MMEA-WI [17]. CPDEA, DNPD, and MMEA-WI perform well in solving MMOP-NON-ICD and MMOP-ICD. With the emergence of MMOPL, the original convergence-weak-relaxed mechanism no longer has advantages. Many scholars have enhanced the degree of relaxation of the convergence-first mechanism and proposed new algorithms based on the convergence-strong-relaxed mechanism, including MMOEA/DC [18], HREA [8], and RHEA [26]. MMOEA/DC, HREA, and RHEA mainly select solutions with better convergence in the local range to cover the global and local PS. Due to the excessive relaxation of the convergence-strong-relaxed mechanism, the selection pressure to improve convergence will be too small, so the convergence will become worse, and even PSs that are difficult to search in the high-dimensional MMOP-ICD are lost. Then, the intensity of interaction between populations in the traditional strong and weak interaction mechanism [15,16] will lead to an imbalance between the diversity and convergence of the decision space, thereby worsening the convergence of the population and even losing PSs that are more difficult to search. Furthermore, existing algorithms that balance the diversity between decision and objective spaces mostly use niche methods. If the niche radius is too large or too small, it is not conducive to the balance of diversity between the two spaces.

As shown in Fig. 5(a), solution A is dominated by solutions C and D in its neighborhood, and solution B is also dominated by solutions C and D in its neighborhood, so solutions A and B have the same local convergence quality. The convergence-strong-relaxed mechanism will select solution A with better diversity while ignoring solution B that is closer to PS_1, resulting in poor convergence of the population. Since many dominance-resistant solutions exist in high-dimensional space, the existing convergence-strong-relaxed mechanism will cause the algorithm to select solutions with better diversity but poor convergence during the environment selection process, so the algorithm cannot search for the true PS. In the co-evolution process, the main population's role is to maintain the convergence of the population. As shown in Fig. 5(a), in the initial stage of evolution, fewer solutions are gathered around PS_2 with greater search difficulty. Suppose the main population and the auxiliary population interact prematurely. In that case, fewer solutions will be distributed near PS_2 in the main population, which is not conducive to the population's exploration and development of PS_2. They may even lead to the loss of PS_2.

Given the above shortcomings, we proposed a new dual-population weak interaction Coevolutionary MMEA called CDP-BCD. As shown in Fig. 4(c), compared with the traditional strong and weak interaction evolution mechanism, CDP-BCD explores and locates the positions of global and local PSs through the auxiliary population Pop 2 before interaction between populations, ensuring the diversity of populations in the decision space and relaxing the convergence-first mechanism. As shown in Fig. 5(b), the main and auxiliary populations can interact when enough solutions are gathered near PS_2. At this time, more solutions in the main population gather around PS_2, which is conducive to the population's exploration and development of PS_2. When updating the auxiliary population, the SLCQ indicator is used. Compared to the local convergence quality in literature [8,26], SLCQ enhances the

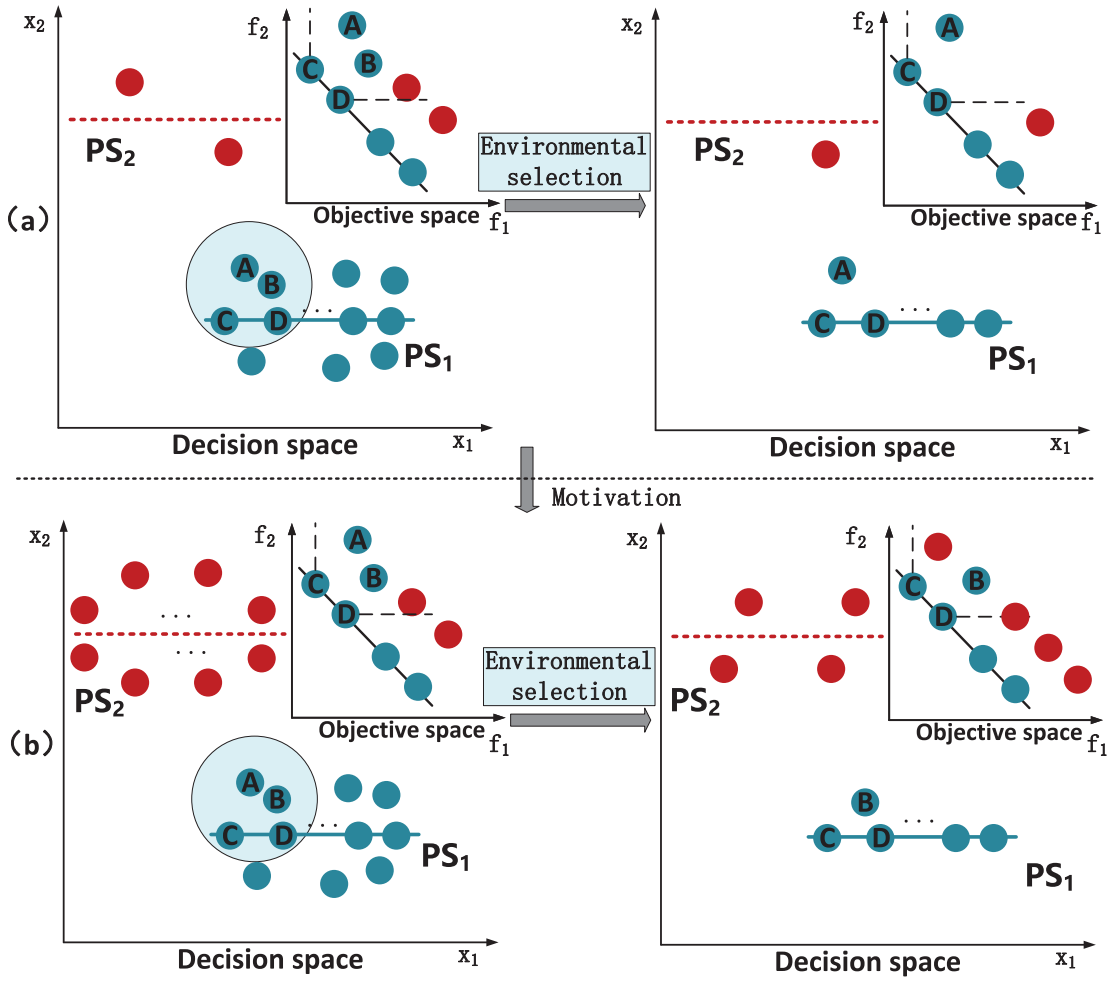


Fig. 5. Introduction to motivation. PS_1 is easier to search, and PS_2 is more difficult to search.

Algorithm 1 Main Framework of CDP-BCD

Input: N (Population size).
Output: $Pop2$ (Final output population).

1: $Pop1 = Initialization(N)$; ▷ The auxiliary population
2: $Pop2 = Initialization(N)$; ▷ The main population
3: $[Pop1, Pop1CD] = UpdateAuxiPop(Pop1, N)$;
4: $[Pop2, Pop2CD] = UpdateMainPop(Pop2, N)$;
5: **while** Termination condition is not met **do**
6: **if** $length(Pop2) < N$ **then**
7: $MPool = TourSel(Pop1, N, Pop1CD)$;
8: **else**
9: $MPool1 = TourSel(Pop1, N/2, Pop1CD)$;
10: $MPool2 = TourSel(Pop2, N/2, Pop2CD)$;
11: $MPool = [MPool1\ MPool2]$;
12: **end if**
13: $Off = OperatorGA(MPool)$;
14: $[Pop1, Pop1CD] = UpdateAuxiPop([Pop1\ Off], N)$;
15: $[Pop2, Pop2CD] = UpdateMainPop([Pop2\ Off], N)$;
16: **end while**
17: Return $Pop2$;

selection pressure to improve convergence and will retain PSs that are more difficult to search in high-dimensional space. SLCQ will retain solution B in Fig. 5(b), which will help improve the convergence of the population. The specific discussion is in Section 5.4. Then, the auxiliary population is used to assist the main population in improving the convergence of the population. Finally, a newly proposed parameter-free distance-based subset selection scheme balances the diversity between the objective and decision spaces.

3. Proposed framework

3.1. Main framework of CDP-BCD

The framework of CDP-BCD is described in Algorithm 1. The main process of CDP-BCD includes population initialization (lines 1, 2), matching selection (lines 7, 9, 10), offspring generation (line 13), updating the auxiliary population (lines 3, 14), and updating the main population (lines 4, 15). The main innovation of CDP-BCD is reflected in the update of the auxiliary population and main population. When updating the auxiliary population, the preference for solutions with good SLCQ makes the population move closer to the true PSs, which helps the algorithm locate multiple equivalent PSs, thereby preventing the loss of imbalanced PSs with greater search difficulty or local PSs, as shown in Fig. 2(b). However, although the solutions in the auxiliary population have good local convergence, their global convergence is poor. Therefore, when updating the main population, we use the niche-based truncation strategy to prune the solutions with good SLCQ. Then, when the size of the pruned population is more excellent than N, a distance-based subset selection method is employed to select N

solutions from this pruned population, which can effectively reduce the diversity loss, as shown in Fig. 2(a).

When the size of the main population is less than N, the algorithm will select the parent from the auxiliary population to generate offspring, so there are enough solutions in the main population that can cover all equivalent PSs in the later stage of evolution. When the size of the main population is equal to N, the algorithm selects $N/2$ parents from the auxiliary population and the main population, respectively, to jointly produce offspring. This is so the auxiliary population can better assist the main population in balancing the convergence and diversity in the decision space.

3.2. Update the auxiliary population

Algorithm 2 Update the Auxiliary Population

Input: $UnionPop$ (Population1 and Offspring), N (Population Size).
Output: Pop (Selected Population), $PopCD$ (Crowding Distance of Pop)

1: $LocalC = CalSLCQ(UnionPop)$;
2: $UnionPopCD = CalCrowdDis(UnionPop)$;
3: $UnionPop = Sortrows(LocalC, UnionPopCD)$;
4: $Pop = UnionPop(1:N)$;
5: $PopCD = CalCrowdDis(Pop)$;
6: Return Pop, $PopCD$;

Algorithm 2 is the primary process of updating the auxiliary population. We first compute the population's SLCQ (line 1). Then the first N solutions with smaller SLCQ and better diversity (lines 2, 3, 4) are retained. Specifically, to update the auxiliary population and main population, we employ the following approach to estimate the density around one solution in the decision space:

$$CrowdDis(x) = \frac{1}{\sigma_x^k + 2}. \tag{2}$$

where σ_i^k is the Euclidean distance between solution x and its kth nearest neighboring solution.

The auxiliary population mainly helps the algorithm locate multiple equivalent PSs. Because convergence is better in multimodal multi-objective optimization when the solution is closer to the equivalent true PS, we design a SLCQ evaluation method. When evaluating the SLCQ of the solution A, we find the neighbors N_A of the solution A in the decision space. If any solution does not dominate solution A in N_A, we call A a local Pareto optimal solution. In addition, in high-dimensional spaces, mainly when the search difficulty differs significantly among PSs, it is not easy to locate all equivalent PSs if only the Pareto domination relationship between neighbors is used to calculate SLCQ. This result is because, in the process of evolution, relatively few solutions are gathered near the imbalanced PS, which is more difficult to search. For these considerations, the algorithm introduces the distance between the solution A and the solutions which dominate A in the neighbors N_A into the calculation of SLCQ. Once there is a solution with better convergence near the PS with more incredible search difficulty, the other solutions closer to this solution will be selected with a high probability, so the imbalanced PS with greater search difficulty will not be discarded.

The SLCQ of the solution x_i is calculated as follows:

$$SLCQ(x_i) = \frac{\sum_{j=1}^{n_i} B_{i,j} d_{i,j}}{n_i R}, \tag{3}$$

while $B_{i,j}$ is 1 if x_i is dominated by x_j and 0 otherwise. $d_{i,j}$ is the Euclidean distance between x_i and x_j in the decision space. n_i is the number of neighbors of the solution x_i. R is the neighborhood radius in the decision space.

If the distance $d_{i,j}$ between x_i and x_j in the decision space is less than radius R, then x_j is said to be a neighbor of x_i. R can be defined by the following:

$$R = \eta \sqrt{\sum_{m=1}^{M} (x_m^{max} - x_m^{min})^2}, \tag{4}$$

where M is the dimension of the decision space. x_m^{max} and x_m^{min} are the maximum and minimum values of the mth dimension in the decision space of the current population, respectively. η is a parameter used to control the size of the neighborhood radius.

Fig. 6 shows an example calculation of SLCQ. First, calculate the neighborhood radius of the current population, such as the radius R of the blue circle in Fig. 6. Take the solution D as an example. D's neighbors are B and E. Therefore, the n_i of D is 2. Since B dominates D in the objective space, the $B_{D,[BE]}$ is [1,0]. Consequently, the SLCQ of D is $0.5 * \frac{d_{DB}}{R}$. In Fig. 6, we can also calculate the SLCQ of E as $0.5 * \frac{d_{EB}}{R}$. Suppose the Euclidean distance in SLCQ is removed and d_{EB} is smaller than d_{DB}, both D and E have a local convergence quality of 0.5. Normally, D will be retained due to its better diversity, while E will be deleted due to its poor diversity. However, D is far from the true PS, so keeping D is not conducive to the positioning of the equivalent PSs by the auxiliary population. Therefore, using SLCQ allows the algorithm to prefer E over D when adding Euclidean distance, thereby enhancing the ability of the auxiliary population to locate the equivalent PS. In addition, when there is a pseudo-local optimal solution around the global PS, the SLCQ value of the pseudo-local optimal solution will be large, and the algorithm will delete it during the selection process.

We prioritize the local Pareto optimal solutions when updating the auxiliary population. As shown in Fig. 2(a), the deleted solutions are deleted because they have poor global convergence. Still, these deleted solutions will be retained to develop their neighboring PSs because they have better SLCQ when updating the auxiliary population. Since SLCQ only represents convergence in a local range, the calculation of SLCQ for solutions near different PSs will not affect each other. Therefore, when encountering an imbalanced PS with greater search difficulty or local PS, as shown in Fig. 2(b), the preference for the solution with smaller SLCQ can make the auxiliary population locate to either the imbalanced PS or local PS.

3.3. Update the main population

When updating the auxiliary population, the algorithm only pays attention to the SLCQ to locate multiple equivalent PSs, which leads to the deterioration of the global convergence and diversity of the population. For this reason, when updating the main population, the algorithm enhances the convergence of the population and balances the convergence and diversity of the population in the decision space. Algorithm 3 is the whole process of updating the main population. Firstly, the solution whose SLCQ is 0 in the union population is reserved (lines 1, 2) in $TmpPop$ to cover all true PSs as much as possible. Only the solutions with an SLCQ value of 0 are kept in the main population instead of the nondominated solutions in the entire union population, thereby relaxing the convergence-first mechanism and balancing the convergence and diversity in the decision space. The nondominated solutions are stored in $TmpPop$ (lines 3, 4) in the set $CPop$, and the niche-based truncation strategy (NBT) is executed for the remaining solutions (lines 6–17) in $TmpPop$. NBT is mainly used to delete the solutions that do not contribute much to the convergence of the population in $remianPop$. When the number of solutions in $CPop$ exceeds N, N solutions are selected in $CPop$ through a distance-based subset selection (DSS) scheme (line 19) to form the final archive set $archive$. The DSS can balance the diversity between the objective and decision spaces.

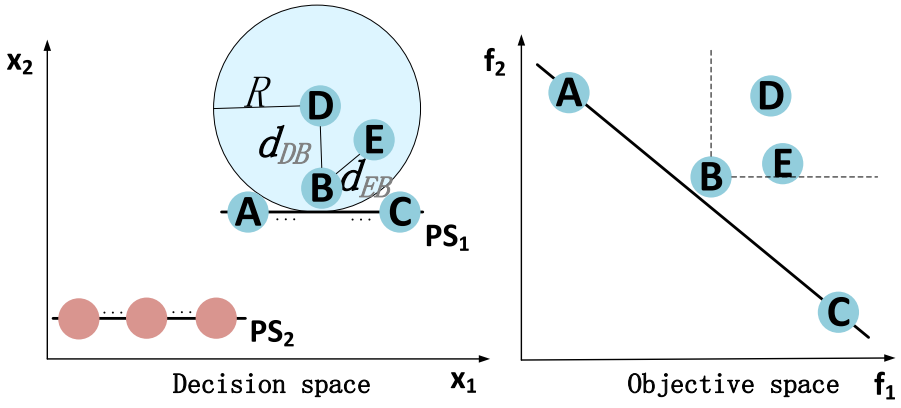


Fig. 6. An example of calculating the strength local convergence quality.

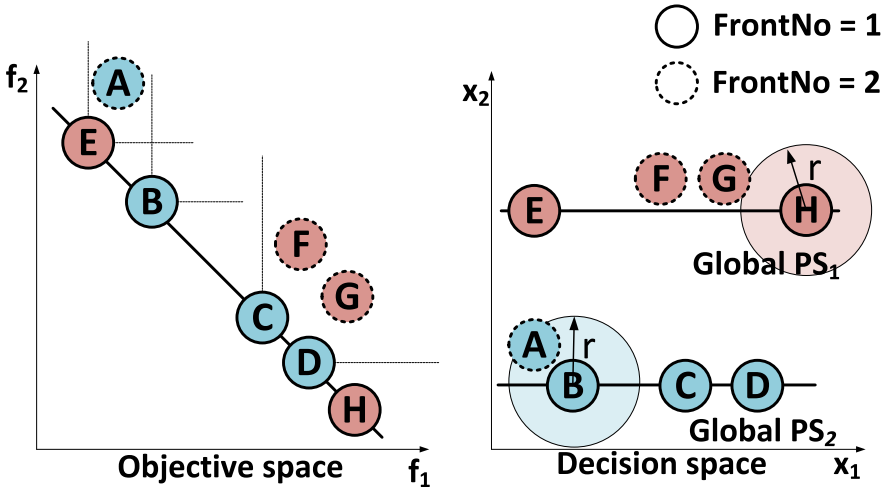


Fig. 7. The role of niche-based truncation strategy.

3.3.1. Niche-based truncation strategy

We propose a niche-based truncation strategy (NBT). Since SLCQ represents the convergence of the solution in a local area in the decision space, there will be many similar solutions with the same SLCQ value in this local area. Computational resources are wasted when a local area is exploited and explored by multiple similar solutions simultaneously. Therefore, NBT uses a niche to filter out and delete the solutions that do not contribute much to the convergence.

Although the solutions with an SLCQ value equal to 0 are closest to the true PS, not all solutions contribute significantly to convergence and even waste computing resources during evolution. For example, as shown in Fig. 7, although the solution A on PS_2 is an optimal local solution, it is globally dominated by the solution E on PS_1. However, in the local area of the solution A, there is a solution B with better convergence to explore and develop it, so the solution B will contribute more than the solution A in the convergence. Although solutions F and G are globally dominated by solutions C and D, respectively, there are no better solutions near solutions F and G to explore and exploit their respective local areas, so these two solutions are best kept. Additionally, solutions in some layers will not contribute to the diversity and convergence of the population, such as some layers composed of a small number of remote solutions in the later stage of evolution. Therefore, when pruning each layer, we delete the layer with a small number of solutions.

$$r = \frac{\sum_{i \in TmpPop} \sum_{l=1}^{L} E_i^l}{L * N}. \tag{5}$$

Before performing NBT, determine a niche radius r from the population waiting to be pruned. The niche radius r represents the average size of the local region in the decision space allocated to the solution to be explored and exploited. The calculation of r is shown in formula (5), E_i^l is the Euclidean distance between solution i and its lth nearest neighboring solution. The NBT approach is straightforward: deleting solutions whose distance to $CPop$ in $RemainPop$ is less than r. As shown in Fig. 7, fast nondominated sorting divides all local optimal solutions into two layers. The algorithm will keep the nondominated solutions in the first layer and then perform the NBT strategy on the solutions in the second layer. Eventually, solutions F and G will be kept, while A will be deleted.

3.3.2. Distance-based subset selection strategy

The distance-based subset selection method (DSS) performs well when dealing with multi/many-objective optimization problems [35], such as hypervolume-based subset selection methods [36,37] and weight-vector-based subset selection methods [11,38]. As described in Algorithm 4, DSS is a greedy-based subset selection method. DSS first finds a solution with the largest crowding distance among the remaining solutions in A as the initial solution of the set P. DSS assigns a distance value D_i to each solution x_i in $A \setminus P$, where D_i represents the minimum distance from x_i to the solution set P. Finally, the solution x_i with the largest D_i in $A \setminus P$ is added to P. It is worth noting that the detailed

Algorithm 3 Update the Main Population

Input: $UnionPop$ (Population2 and Offspring), N (Population Size).
Output: Pop (Selected Population), $PopCD$ (Crowding Distance).

$LocalC = CalSLCQ(UnionPop)$;
$TmpPop = UnionPop(LocalC == 0)$;
$[FrontNo, \ ] = NDSort(TmpPop)$;
$CPop = Tmp(FrontNo == 1)$;
$RemainPop = Tmp(FrontNo \neq 1)$;
while $isempty(RemainPop)$ **do** ▷ Niche-based Truncation
  $RemainPop = NBT(RemainPop, CPop)$;
  $[FNoR, MaxFNo] = NDSort(RemainPop)$;
  $i = \max_{i=1...MaxFNo}(|FNoR == i|)$;
  $SPop = RemainPop(FNoR == i)$;
  if $len(SPop) < delta$ and $gen > maxFE * 0.5$ **then**
    $break$;
  else
    $CPop = CPop \cup SPop$;
    $RemainPop = RemainPop \setminus SPop$;
  end if
end while
if $len(CPop) > N$ **then** ▷ DSS
  $Archive = DSS(CPop, N)$;
else
  $Archive = CPop$;
end if
$ArchiveCD = CalCrowdDis(Archive)$;
Return $Archive, ArchiveCD$;

Algorithm 4 Distance-Based Subset Selection

Input: A (All evaluated solutions), N (The number of solutions to be selected).
Output: P (Selected subset).

$A \leftarrow$ sort A based on crowding distance;
$P \leftarrow$ the first solution in the sorted set A;
while $|P| < N - 1$ **do**
  for each $x_i \in A \setminus P$ **do**
    $D_i \leftarrow$ minimum Euclidean distance from x_i to P: $D_i = \min\|F(x_i) - F(x_j)\| + \|x_i - x_j\| \ \forall j \in 1, ...|P|$;
  end for
  Choose solution from $A \setminus P$ associated with the largest Euclidean distance and add it to P;
end while
Return P;

definition of D_i is as follows:

$$D_i = \arg\min_{x_j \in |P|} d(x_i, x_j) = \arg\min_{x_j \in |P|} d_{obj}(x_i, x_j) + d_{dec}(x_i, x_j), \tag{6}$$

where $d_{obj}(x_i, x_j)$ and $d_{dec}(x_i, x_j)$ represent the distances of x_i and x_j in the objective space and decision space, respectively.

As shown in Fig. 8, if the traditional convergence-first strategy is used in environmental selection, the algorithm will delete the solution set (H, I, J). Then, it will directly select six solutions from the solution set (A, B, C, D, E, F, G) according to the degree of congestion. Although (H, I, J) have poor global convergence, they are beneficial to the exploitation of $PS1$, maintaining the convergence and diversity of PS_1. When the NBT is implemented, those solutions that do not contribute significantly to the convergence are deleted, followed by an improvement in the population diversity.

Applying DSS can also help the algorithm find more true PSs. For example, if the initial solution is on one PS, the following solution selected by DSS is highly likely to be on the other PSs. By weighing distance in the selection process, the population can cover more true PS. In Fig. 8, the solution A with the slightest degree of crowding is first used as the initial solution of DSS. Then according to the maximum and minimum distance formula in line 5 of Algorithm 4, the solution J is selected. And the solutions (F, E, C, I) are selected by DSS in turn. Therefore, DSS enables the solutions to be evenly distributed over all PSs.

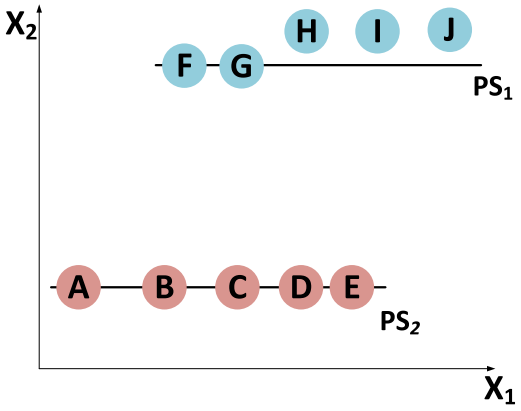


Fig. 8. The Selection Criteria of DSS. Six solutions need to be selected.

3.4. Principles of CDP-BCD

To demonstrate how CDP-BCD works, we list a simple example for solving an MMOP with only the global PS in Fig. 9(a)–(d). Fig. 9(a) is the initial population, with 13 solutions, and six solutions must be selected as the next-generation solution set. CDP-BCD first uses SLCQ to locate the approximate positions of all PSs when updating the auxiliary population. Fig. 9(b) shows the solutions in the auxiliary population, solution set (1, 2, 3, 6) and solution set (7, 8, 9, 10, 13) are clustered around two true equivalent PSs. Fig. 9(c) shows the population pruned by NBT. NBT will delete solutions that do not contribute much to the convergence of the population, such as solution 10 and solution 6, because there are better convergent solutions around solution 10 and solution 6 to explore and develop their respective neighborhoods. Finally, DSS selects the final population as shown in Fig. 9(d).

3.5. Complexity analysis

The main steps of CDP-BCD include updating the auxiliary and main populations. The calculation of SLCQ is mainly involved in updating the auxiliary population. The computational complexity of SLCQ is $\mathcal{O}(N^2)$. The calculation of updating the main population is mainly consumed in the calculation of the SLCQ and DSS operations, and the computational complexity of the DSS is $\mathcal{O}(N^2)$, so the computational complexity of updating the main population is $\mathcal{O}(N^2+N^2)$. Finally, the computational complexity of CDP-BCD is $\mathcal{O}(N^2 + N^2 + N^2) = \mathcal{O}(N^2)$.

4. Experimental design

4.1. Benchmark problems

In this study, we used different types of benchmark problems to verify the effectiveness of CDP-BCD in solving MMOPs. Yue et al. [6] proposed multimodal multi-objective test functions (MMFs). In the MMFs, MMF1-MMF9, MMF14, and MMF14_a contain only global PSs, while MM10-MMF13, MMF15, and MMF15_a have a local PS. The PSs of MMF1-MMF15 have complex manifolds, especially the manifolds of PSs of MMF1, MMF5, and MMF6 are complex sin functions. Moreover, the equivalent PSs of some MMFs overlap in some dimensions, which undoubtedly brings great challenges to the algorithm design.

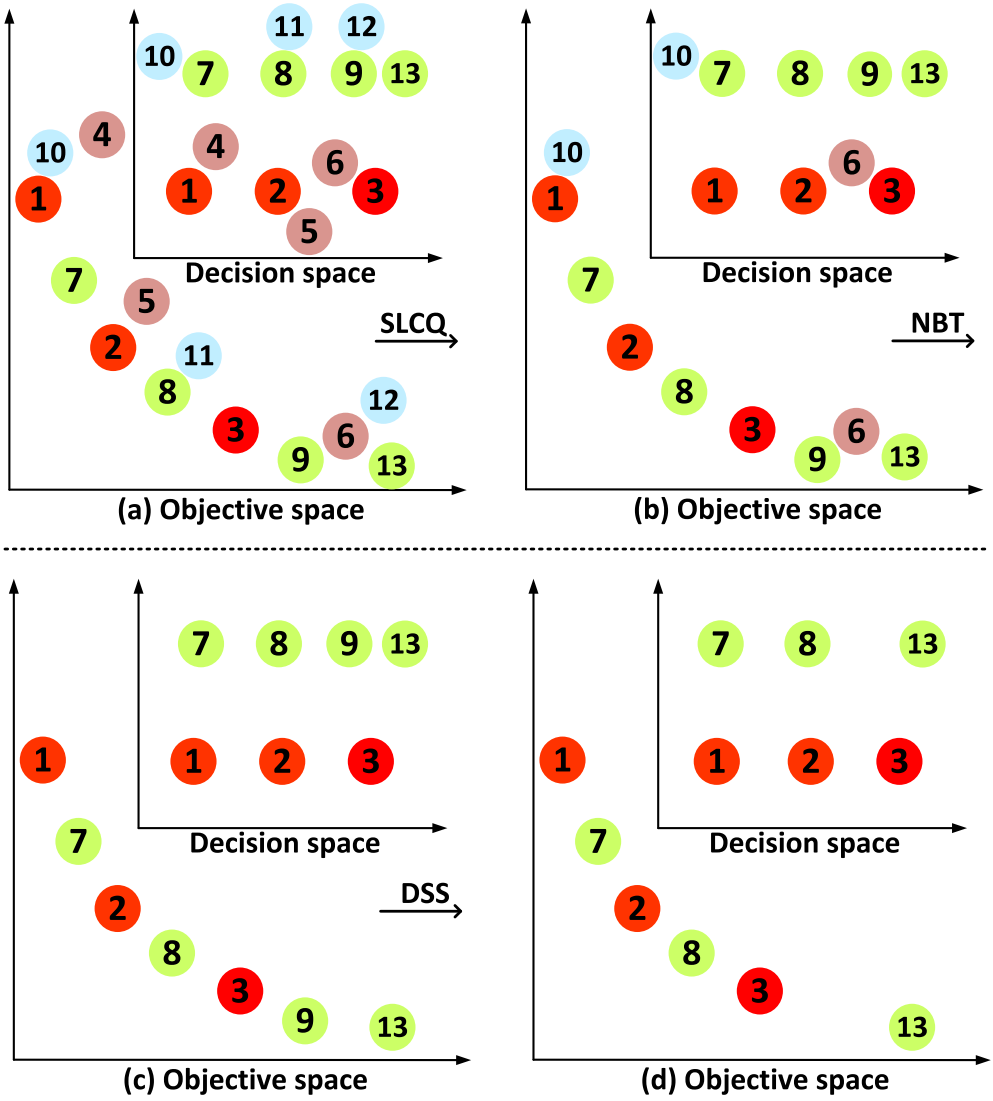


Fig. 9. Principles of CDP-BCD.

Polygon-M Problem [39] with local PSs was proposed by Liu et al. M represents the dimension of the objective space. The dimensionality of the objective space of Polygon-M is easily extended to high-dimensional spaces.

The search difficulty for each PS of the MMOP just described is similar. Liu et al. [7] proposed the imbalanced distance minimization problems (IDMPs) with different search difficulties for different PSs. IDMP is derived from distance minimization problems [40] and polygon-based problems [41]. In addition, Li et al. [8] proposed the imbalanced distance minimization benchmark problem with epsilon-efficient solutions (IDMP_e). The IDMP_e is a variant of the IDMP. The local PF of the IDMP_e is mapped by multiple equivalent local PSs, and the global PF of the IDMP_e is close to the local PF. The aforementioned IDMP and its variants all belong to MMOP-ICD.

For all benchmark problems, the population size of the algorithm at execution time is set at D*100, and the total number of evaluations is set D*10000. D is the dimension of the decision space.

4.2. Parameter settings

To verify the effectiveness of the proposed CDP-BCD, we chose eight state-of-the-art MMEAs as comparative algorithms. They are MO_Ring_PSO_SCD [14], TriMOEA-TA&R [15], CPDEA [7], MMO-ClusteringPSO [42], MMOEA/DC [18], MMEA-WI [17], RHEA [26], and HREA [8] which are reviewed in Section 2. The parameter settings of all the comparison algorithms are shown in Table 1. The GA operator uses simulated binary crossover (SBX) [43] and polynomial mutation (PM) [44]. Dis_c is the distribution index of the crossover, and Dis_m is the distribution index of the mutation. P_C and P_M are the crossover probability and mutation probability, respectively.

4.3. Performance indicator

Traditional performance evaluation indicators usually measure the comprehensive performance of the objective space, such as GD [45], IGD [46], and HV [47], which can measure the convergence and diversity of the population in the objective space. Nevertheless, for MMOPs, the performance evaluation is limited to the objective space and should also be extended to the decision space. Therefore, in this study, Inverted Generational Distance-Multimodal (IGDM) [15] and averaged Inverted Generational Distance in the decision space (IGDX) [48] were used as indicators to evaluate the performance of the proposed algorithm.

IGDM can measure convergence and diversity performance in the objective and decision spaces. A more uniform and widely distributed

Table 1
The parameter settings of the comparison algorithms.

Algorithm	Parameters
MO_Ring_PSO_SCD	$C_1 = C_2 = 2.05$, $W = 0.7298$, $n_{PBA} = 5$, $n_{NBA} = 15$
TriMOEA-TA&R	$p_{con} = 0$, $\sigma_{niche} = 0.5$, $\epsilon_{peak} = 0.01$
CPDEA	$\eta = 2$, $K = 3$
MMO-ClusteringPSO	$C_1 = C_2 = 2.05$, $W = 0.7298$, $n_{PBA} = 5$, $n_{NBA} = 15$, $subsize = 5$, $n_{GBA} = 10 * subsize$
MMOEA/DC	$Dis_c = Dis_m = 20$, $P_C = 1$, $P_M = 1/\text{n}$, $\lambda = 0.1$, $\beta = 0.5$
MMEA-WI	$p = 0.4$, $\kappa = 0.05$
RHEA	$Dis_c = Dis_m = 20$, $P_M = 1/\text{n}$
HREA	$p = 0.5$, $\epsilon = 0.3$
CDP-BCD	$Dis_c = Dis_m = 20$, $P_C = 1$, $k = \sqrt{N}$, $\eta = 0.2$, $L = 3$

solution set in the objective space will have smaller IGDM values. In addition, the more Pareto optimal solutions (or solutions very close to them) found in the decision space for a given PF, the smaller the IGDM value will be.

Let $F^* : \{f_1^*, f_2^*, \ldots, f_q^*\}$ be the set of well distributed reference points (in the objective space) on the PF, where q is the number of reference points. Let A be $a_1, a_2, \ldots, a_q$, where $a_i, i = 1, 2, \ldots, q$ is the number of Pareto optimal solutions for each point, f_i^*. Let $X^* : \{x_{1,1}^*, x_{1,2}^*, x_{1,a_1}^*, x_{2,1}^*, \ldots, x_{q,a_q}^*\}$ be the set of Pareto optimal solutions (in the decision space) corresponding to F^* and A. Let $P : \{x_1, x_2, \ldots, x_p\}$ be the approximate solution set obtained by an optimizer, where p is the number of the obtained solutions. Metric IGDM is defined as follows:

$$IGDM(P, F^*, X^*) = \frac{\sum_{f_i^* \in F^*} \sum_{j=1,\ldots,a_i} d(f_i^*, x_{i,j}^*, P)}{|X^*|},$$

$$d(f_i^*, x_{i,j}^*, P) = \begin{cases} d_{max}, if \ \ P_{i,j} = \emptyset, \\ \min\{d_{max}, ed(f_i^*, P_{i,j})\}, else, \end{cases}$$

$$P_{i,j} = \{x_k : j = \arg\min_{l=1,\ldots,a_i} ed(x_{i,l}^*, x_k), x_k \in P\}, \tag{7}$$

where $ed(f_i^*, P_{i,j})$ is the minimal Euclidean distance between f_i^* and $P_{i,j}$, $ed(x_{i,l}^*, x_k)$ is the Euclidean distance between $x_{i,l}^*$ and x_k, and d_max is a parameter defined by the user.

IGDX is a variant of IGD, and IGDX values represent the average distance (Euclidean distance) between the obtained solutions and reference solutions (true PS) in the decision space. It can evaluate the diversity and convergence in the decision space. We can calculate it as follows:

$$IGDX(O, P^*) = \frac{\sum_{v \in P^*} d(v, O)}{|P^*|}, \tag{8}$$

where P^* denotes a set of uniformly distributed points along the true PS; O denotes a set of obtained solutions, and $d(v, O)$ is the minimum Euclidean distance between v and the points in O. Smaller IGDX value means better diversity and convergence in the decision space.

5. Experimental results

This section mainly analyzes the experimental results on different indicators. The IGDX indicator measures the convergence and diversity of the comparison algorithms in the decision space. IGDM extends IGDX to objective space, which can measure the convergence and diversity of the decision and objective spaces at the same time. Furthermore, we discussed the various components and parameters of the CDP-BCD. Afterward, experimental results were obtained by independently running the algorithms 30 times on each benchmark problem. Finally, to obtain comprehensive statistical conclusions, the Wilcoxon rank-sum test [49] was used to compare the results

obtained by the CDP-BCD and the other algorithms at a significance level of 0.05. Symbols "+ ", "−", and "≈" indicate the compared algorithm is significantly better than, significantly worse than, or statistically tied by CDP-BCD.

5.1. Comparative results on the MMOP-ICD and polygon-based MMOPL

2 is the IGDX indicator result of nine algorithms on the MMOP-ICD and Polygon-based MMOPL. CDP-BCD performed best on 20 benchmark problems. HREA has the best IGDX value on seven benchmark problems. RHEA has the best IGDX value on one benchmark problem. The results of HREA on IDMPM3T1, IDMPM3T2, IDMPM3T3, and IDMPM3T4 are better than CDP-BCD. The hierarchical sorting method in HREA can help the algorithm to separate imbalanced PSs or local PSs from PSs that are easier to search for and then make HREA perform better in the decision space by controlling the number of solutions on each equivalent PS. However, due to the instability of the local convergence quality in HREA (see Section 5.4 for details), the performance of HREA is poor when there are more decision variables or the difference in search difficulty between equivalent PSs becomes larger in some MMOPs, so HREA performed worse than CDP-BCD on IDMPM4T1, IDMPM4T2, IDMPM4T3 and IDMPM4T4. Due to the long distance between the global PF and local PF in the Polygon-based MMOPL, the hierarchical sorting method in HREA deletes the solution on the local PF, so the effect of HREA on the Polygon-based MMOPL is poor. Although CDP-BCD performed better than RHEA on most benchmark problems, the gap between CDP-BCD and RHEA on IGDX is smaller on the Polygon-based MMOPL. Moreover, MMOEA/DC also performed well on most MMOP-ICD and Polygon-based MMOPL. CDP-BCD outperformed MO_Ring_PSO_SCD, TriMOEA-TA&R, CDPEA, MMO_Clustering_PSO and MMEA-WI on all benchmark problems. Therefore, CDP-BCD performed well in the decision space when solving MMOP-ICD and Polygon-based MMOPL.

IGDM can simultaneously evaluate the convergence and diversity of the population in the decision and objective spaces. Table 3 is the IGDM indicator result on the MMOP-ICD and Polygon-based MMOPL. CDP-BCD has the best IGDM value on the 13 benchmark problems. HREA and RHEA have seven and six best IGDM values, respectively. MMOEA/DC and MMEA-WI have the best IGDM value on one benchmark problem. Since HREA only pays attention to the diversity of the decision space, the performance of HREA was worse when the diversity between the decision and objective spaces was imbalanced. While CDP-BCD, RHEA, and MMOEA/DC have mechanisms to balance the diversity between the decision and objective spaces, these three algorithms have better IGDM results. Particularly RHEA, on the Polygon-based MMOPL, is slightly better than CDP-BCD on the IGDM performance indicator. Although MMOEA/DC has only one best IGDM value, compared with other algorithms except for CDP-BCD, it is also very competitive in the IGDM performance indicator. Since MO_Ring_PSO_SCD, TriMOEA-TA&R, and MMO_Clustering_PSO themselves are very difficult when solving MMOP-ICD and Polygon-based MMOPL, these three algorithms have poor IGDM values on the MMOP-ICD and Polygon-based MMOPL. In summary, CDP-BCD also has better performance in the objective space.

It is worth noting that on the Polygon-based MMOPL, the performance of CDP-BCD on IGDX is different from that on IGDM. Since the convergence properties between the decision and objective spaces are the same, the difference mentioned above is caused by the gap

Table 2
IGDX results obtained by compared algorithms on the MMOP-ICD and Polygon-based MMOPL.

Problem	MO_PSO_Ring_SCD	TriMOEA-TA&R	CPDEA	MMO_Clust_ering_PSO	MMOEA/DC	MMEA-WI	RHEA	HREA	CDP-BCD
IDMPM2T1	1.6804e−1	6.7326e−1	6.5216e−4	2.0591e−3	6.1474e−4	8.2852e−4	1.0392e−3	6.4824e−4	**5.7967e−4**
	(2.84e−1) −	(1.88e−5) −	(4.60e−5) −	(3.90e−4) −	(1.88e−5) −	(4.89e−5) −	(6.51e−4) −	(5.30e−4) −	**(1.96e−5)**
IDMPM2T2	4.5322e−3	4.0653e−1	7.6574e−4	1.3438e−3	8.7087e−4	1.0344e−3	1.3989e−3	8.1187e−4	**7.2895e−4**
	(2.96e−3) −	(3.32e−1) −	(8.14e−5) =	(2.36e−4) −	(5.93e−5) −	(5.81e−5) −	(7.45e−4) −	(4.74e−5) −	**(4.95e−5)**
IDMPM2T3	2.2675e−3	2.2456e−1	1.5853e−3	1.4342e−3	1.3389e−3	1.4431e−3	4.1678e−3	1.3438e−3	**1.1439e−3**
	(2.46e−4) −	(3.00e−1) −	(2.26e−4) −	(1.25e−4) −	(2.27e−4) −	(1.05e−4) −	(4.33e−3) −	(1.70e−4) −	**(1.01e−4)**
IDMPM2T4	6.3101e−3	6.0601e−1	6.3921e−4	1.4641e−3	1.5757e−1	4.5745e−2	2.1627e−3	2.1789e−3	**5.7827e−4**
	(4.83e−3) −	(2.05e−1) −	(3.76e−5) −	(3.83e−4) −	(2.89e−1) −	(1.71e−1) −	(1.86e−3) −	(5.34e−3) −	**(1.34e−5)**
IDMPM3T1	9.4341e−2	7.4500e−1	6.7562e−3	1.0272e−2	7.6470e−3	7.5195e−3	7.4631e−3	**6.4308e−3**	6.7207e−3
	(1.13e−1) −	(2.69e−1) −	(7.19e−5) =	(1.14e−3) −	(1.06e−4) −	(1.90e−4) −	(4.25e−4) −	**(8.15e−5)** +	(1.03e−4)
IDMPM3T2	5.1222e−2	5.7902e−1	6.7816e−3	9.9040e−3	7.6810e−3	7.6817e−3	7.7243e−3	**6.4991e−3**	6.7199e−3
	(8.38e−2) −	(2.88e−1) −	(8.14e−5) −	(9.93e−4) −	(1.33e−4) −	(1.83e−4) −	(6.40e−4) −	**(6.72e−5)** +	(1.19e−4)
IDMPM3T3	1.2379e−2	4.1357e−1	7.8050e−3	9.0110e−3	8.6023e−3	8.1825e−3	1.2973e−2	**7.0681e−3**	7.2289e−3
	(9.81e−4) −	(2.87e−1) −	(2.01e−4) −	(2.59e−4) −	(1.69e−4) −	(2.69e−4) −	(6.27e−3) −	**(8.97e−5)** +	(1.00e−4)
IDMPM3T4	5.5424e−2	8.2065e−1	6.7090e−3	1.3704e−2	2.5991e−2	4.8252e−2	9.6222e−3	**6.3734e−3**	6.6282e−3
	(8.39e−2) −	(2.59e−1) −	(7.81e−5) −	(3.22e−3) −	(5.28e−2) −	(9.25e−2) −	(3.01e−3) −	**(6.21e−5)** +	(9.32e−5)
IDMPM4T1	6.0014e−1	1.1775e+0	2.0140e−1	3.4197e−2	1.7553e−2	7.5493e−3	1.8192e−2	2.4623e−2	**7.0812e−3**
	(2.37e−1) −	(8.07e−2) −	(1.55e−1) −	(4.79e−2) −	(4.75e−2) −	(1.53e−4) −	(6.09e−3) −	(1.08e−2) −	**(1.66e−4)**
IDMPM4T2	3.2263e−1	1.0185e+0	2.8194e−1	5.1836e−2	2.6531e−2	7.7897e−2	3.1220e−2	7.8539e−2	**7.5594e−3**
	(2.11e−1) −	(2.13e−1) −	(1.97e−1) −	(7.84e−2) −	(5.85e−2) −	(1.18e−1) −	(1.03e−2) −	(1.61e−1) −	**(7.83e−4)**
IDMPM4T3	2.1259e−2	8.1603e−1	1.3521e−1	1.6833e−2	1.9929e−2	9.4269e−2	5.3562e−2	1.8275e−1	**8.2730e−3**
	(2.44e−3) −	(2.97e−1) −	(1.32e−1) −	(1.89e−3) −	(4.78e−2) −	(1.61e−1) −	(2.20e−2) −	(2.07e−1) −	**(5.56e−4)**
IDMPM4T4	1.4200e−1	1.1040e+0	1.6610e−1	4.4674e−2	2.2290e−2	3.5775e−1	3.1267e−2	1.6867e−1	**7.2723e−3**
	(1.44e−1) −	(1.41e−1) −	(1.65e−1) −	(6.90e−2) −	(5.30e−2) −	(1.70e−1) −	(1.30e−2) −	(1.76e−1) −	**(1.64e−4)**
IDMPM2T1_e	6.7357e−1	6.7326e−1	6.7320e−1	6.7371e−1	5.6039e−4	6.7325e−1	1.3585e−3	**5.4352e−4**	5.8372e−4
	(5.07e−4) −	(1.46e−5) −	(7.66e−5) −	(5.33e−4) −	(1.82e−5) +	(9.67e−5) −	(5.83e−4) −	**(2.34e−5)** +	(1.61e−5)
IDMPM2T2_e	6.7346e−1	6.7327e−1	6.7324e−1	6.7338e−1	9.6661e−4	6.7332e−1	1.7945e−3	9.2983e−4	**8.3806e−4**
	(5.29e−4) −	(3.42e−5) −	(1.52e−4) −	(3.69e−4) −	(5.26e−5) −	(2.70e−4) −	(7.65e−4) −	(4.65e−5) −	**(8.26e−5)**
IDMPM2T3_e	1.8590e−1	3.5008e−1	2.2550e−1	2.2828e−1	9.3195e−4	2.2565e−1	1.8777e−3	8.7107e−4	**8.5949e−4**
	(1.08e−1) −	(1.33e−1) −	(1.58e−4) −	(3.39e−3) −	(3.13e−5) −	(1.46e−4) −	(1.03e−3) −	(2.12e−5) =	**(2.82e−5)**
IDMPM2T4_e	5.8787e−1	5.7659e−1	5.8801e−1	5.8793e−1	8.7843e−2	5.8803e−1	3.9522e−2	**1.3206e−3**	4.6705e−2
	(6.29e−4) −	(6.08e−2) −	(8.79e−5) −	(4.01e−4) −	(5.64e−2) −	(1.34e−4) −	(1.57e−2) +	**(1.08e−4)** +	(2.25e−2)
IDMPM3T1_e	6.3490e−1	7.7618e−1	6.2471e−1	6.2740e−1	7.7571e−3	6.2481e−1	7.2284e−3	**6.6843e−3**	6.9909e−3
	(5.59e−3) −	(1.87e−1) −	(8.50e−4) −	(1.73e−3) −	(1.67e−4) −	(6.77e−4) −	(1.57e−4) −	**(7.30e−5)** +	(9.06e−5)
IDMPM3T2_e	3.3538e−1	5.3613e−1	3.3045e−1	3.3218e−1	5.2714e−3	3.3061e−1	5.0373e−3	1.6935e−1	**4.8103e−3**
	(3.14e−3) −	(1.12e−1) −	(2.62e−4) −	(1.53e−3) −	(1.09e−4) −	(6.53e−4) −	(3.06e−4) −	(3.96e−4) −	**(7.78e−5)**
IDMPM3T3_e	3.3472e−1	4.7564e−1	3.3236e−1	3.3228e−1	7.6607e−3	3.3230e−1	8.4928e−3	1.6863e−1	**5.2408e−3**
	(2.51e−3) −	(1.64e−1) −	(6.57e−4) −	(1.21e−3) −	(1.08e−2) −	(8.97e−4) −	(1.98e−3) −	(9.09e−4) −	**(9.18e−5)**
IDMPM3T4_e	5.6974e−1	6.5913e−1	5.6590e−1	5.6834e−1	9.3360e−3	6.2322e−1	7.5629e−3	3.3527e−1	**7.1050e−3**
	(4.46e−3) −	(1.03e−1) −	(6.11e−4) −	(3.72e−3) −	(9.19e−3) −	(1.06e−1) −	(1.82e−4) −	(6.42e−4) −	**(1.40e−4)**
Polygon-2	6.3650e−1	6.7431e−1	3.5790e−1	3.0759e−1	8.4399e−4	2.7139e−1	**8.3027e−4**	4.2210e−1	8.4285e−4
	(1.46e−1) −	(1.26e−5) −	(3.02e−1) −	(2.84e−1) −	(2.97e−5) =	(2.90e−1) −	**(2.35e−5)** =	(2.93e−1) −	(3.41e−5)
Polygon-3	5.7348e−1	7.8526e−1	3.0719e−1	4.5432e−1	2.5223e−2	7.3975e−1	1.8134e−2	3.4024e−1	**1.7666e−2**
	(1.64e−1) −	(4.26e−2) −	(1.32e−1) −	(1.53e−1) −	(1.58e−2) −	(1.19e−1) −	(3.23e−4) −	(1.61e−1) −	**(3.75e−4)**
Polygon-4	5.0252e−1	7.8845e−1	2.2297e−1	5.1384e−1	2.2884e−2	7.7995e−1	2.0587e−2	3.0186e−1	**2.0455e−2**
	(1.70e−1) −	(4.37e−2) −	(1.16e−1) −	(2.02e−1) −	(6.66e−3) −	(5.31e−2) −	(4.04e−4) =	(9.81e−2) −	**(4.91e−4)**
Polygon-5	3.8673e−1	7.9103e−1	2.0298e−1	3.3556e−1	2.0330e−2	7.8948e−1	1.9202e−2	1.8492e−1	**1.8967e−2**
	(1.50e−1) −	(4.18e−2) −	(1.14e−1) −	(1.53e−1) −	(1.05e−3) −	(1.45e−3) −	(2.55e−4) −	(9.65e−2) −	**(3.15e−4)**
Polygon-6	3.9893e−1	7.8925e−1	1.6612e−1	3.1405e−1	2.3702e−2	7.4228e−1	1.9652e−2	1.6348e−1	**1.9498e−2**
	(1.60e−1) −	(2.80e−4) −	(8.17e−2) −	(1.29e−1) −	(1.09e−2) −	(9.81e−2) −	(3.17e−4) =	(6.73e−2) −	**(3.48e−4)**
Polygon-8	3.3196e−1	7.7155e−1	1.1783e−1	2.3915e−1	2.2667e−2	7.1664e−1	2.0139e−2	1.3949e−1	**1.9991e−2**
	(1.31e−1) −	(7.91e−2) −	(6.01e−2) −	(1.05e−1) −	(6.75e−3) −	(1.30e−1) −	(2.83e−4) −	(5.49e−2) −	**(2.93e−4)**
Polygon-10	2.8068e−1	7.8590e−1	9.4561e−2	2.1275e−1	2.3239e−2	7.2763e−1	2.0389e−2	1.3165e−1	**2.0315e−2**
	(1.26e−1) −	(6.94e−2) −	(3.25e−2) −	(1.17e−1) −	(4.10e−3) −	(1.21e−1) −	(2.72e−4) =	(5.51e−2) −	**(3.25e−4)**
Polygon-12	2.3470e−1	8.0120e−1	1.0607e−1	1.9118e−1	2.1870e−2	6.6174e−1	2.0495e−2	1.2661e−1	**2.0308e−2**
	(1.10e−1) −	(6.05e−3) −	(5.44e−2) −	(8.00e−2) −	(7.95e−4) −	(1.74e−1) −	(2.62e−4) −	(7.48e−2) −	**(2.79e−4)**
+/−/=	0/28/0	0/28/0	0/26/2	0/28/0	1/26/1	0/28/0	1/23/4	7/20/1	

between the diversity of the decision space and the objective space. In the Polygon-based MMOPL, the distance between two PSs is generally more significant than between two PFs, resulting in a relatively large diversity gap between the decision and objective spaces. RHEA's dual-crowding strategy makes the diversity of the decision space equally important as the diversity of the objective space. In contrast, CDP-BCD's DSS makes the diversity of the decision space slightly more important than the diversity of the objective space. Therefore, on the Polygon-based MMOPL, the IGDX performance of CDP-BCD is better than that of RHEA, while the IGDM performance of CDP-BCD is worse than that of RHEA. Different results also indirectly show that RHEA will perform better when balancing the diversity between the decision and objective spaces. For MMOP-ICD, since CDP-BCD has a great advantage in convergence, there is no significant gap between the performance of CDP-BCD on IGDX and IGDM. However, on IDMPM3Ts, CDP-BCD performs worse than HREA on IGDX and IGDM. This difference in performance is mainly due to the different mechanisms of HREA and CDP-BCD to maintain diversity. DSS considers that the population can evenly cover the true PS under the global situation. In contrast, the diversity maintenance mechanism of HREA mainly considers the distance between neighbors within the niche. As shown in Fig. 8, DSS selected solution C. Still, if the diversity maintenance mechanism of HREA is used, since the crowding degree of solution B is less than C, HREA will choose solution B to enhance the diversity of the population. It can be seen from Fig. 8 that solution B contributes more to diversity than C. Therefore, the performance of CDP-BCD on IDMPM3Ts on IGDX and IGDM is worse than HREA.

Table 3
IGDM results obtained by compared algorithms on the MMOP-ICD and Polygon-based MMOPL.

Problem	MO_PSO_Ring_SCD	TriMOEA-TA&R	CPDEA	MMO_Clustering_PSO	MMOEA/DC	MMEA-WI	RHEA	HREA	CDP-BCD
IDMPM2T1	1.9375e−1	5.0168e−1	4.8056e−3	1.5263e−2	4.5513e−3	5.9454e−3	1.1049e−2	4.6777e−3	**4.1688e−3**
	(1.85e−1) −	(2.03e−5) −	(3.95e−4) −	(2.73e−3) −	(1.99e−4) −	(3.47e−4) −	(9.70e−3) −	(3.74e−3) −	**(1.85e−4)**
IDMPM2T2	2.9959e−2	3.1897e−1	4.4379e−3	7.6201e−3	4.5099e−3	6.0367e−3	9.5362e−3	4.1876e−3	**4.1043e−3**
	(2.09e−2) −	(2.34e−1) −	(1.96e−4) −	(1.68e−3) −	(1.65e−4) −	(3.55e−4) −	(7.44e−3) −	(1.30e−4) −	**(7.97e−5)**
IDMPM2T3	1.1534e−2	2.9791e−1	8.0588e−3	6.4662e−3	5.3810e−3	6.7770e−3	2.2472e−2	6.0984e−3	**5.2414e−3**
	(1.66e−3) −	(2.03e−1) −	(1.61e−3) −	(9.01e−4) −	(6.58e−4) =	(5.22e−4) −	(2.56e−2) −	(1.12e−3) −	**(6.99e−4)**
IDMPM2T4	4.4804e−2	4.5207e−1	4.5056e−3	1.0515e−2	1.2065e−1	3.9540e−2	3.6141e−2	1.5377e−2	**4.0579e−3**
	(3.41e−2) −	(1.51e−1) −	(2.85e−4) −	(2.69e−3) −	(2.14e−1) −	(1.26e−1) −	(4.33e−2) −	(3.78e−2) −	**(1.06e−4)**
IDMPM3T1	2.5767e−1	6.1904e−1	4.8902e−2	7.9500e−2	5.6969e−2	5.3745e−2	5.8508e−2	**4.6543e−2**	4.8148e−2
	(9.20e−2) −	(1.48e−1) −	(6.70e−4) −	(8.85e−3) −	(9.38e−4) −	(1.41e−3) −	(6.42e−3) −	**(7.99e−4)** +	(7.23e−4)
IDMPM3T2	1.6168e−1	5.1770e−1	4.7955e−2	7.0927e−2	5.5251e−2	5.3892e−2	5.7622e−2	**4.5672e−2**	4.7027e−2
	(6.69e−2) −	(1.68e−1) −	(6.08e−4) −	(7.25e−3) −	(1.35e−3) −	(1.37e−3) −	(6.57e−3) −	**(4.05e−4)** +	(8.46e−4)
IDMPM3T3	8.4812e−2	4.4285e−1	5.1806e−2	6.0368e−2	5.8462e−2	5.4764e−2	9.2897e−2	**4.6852e−2**	4.7770e−2
	(7.10e−3) −	(1.78e−1) −	(1.24e−3) −	(1.81e−3) −	(1.36e−3) −	(1.90e−3) −	(4.49e−2) −	**(5.58e−4)** +	(6.44e−4)
IDMPM3T4	1.9903e−1	6.6236e−1	4.8173e−2	1.0188e−1	8.6848e−2	9.2657e−2	7.7424e−2	**4.5737e−2**	4.7019e−2
	(8.17e−2) −	(1.48e−1) −	(8.36e−4) −	(2.45e−2) −	(7.89e−2) −	(8.75e−2) −	(2.56e−2) −	**(6.31e−4)** +	(8.98e−4)
IDMPM4T1	6.0044e−1	7.7865e−1	2.2461e−1	2.1258e−1	9.1603e−2	**5.6586e−2**	1.9571e−1	1.8349e−1	5.7035e−2
	(8.59e−2) −	(2.95e−2) −	(1.34e−1) −	(4.80e−2) −	(3.87e−2) −	**(1.39e−3)** =	(5.40e−2) −	(7.59e−2) −	(2.82e−3)
IDMPM4T2	4.2713e−1	7.2882e−1	2.7527e−1	2.0704e−1	9.1377e−2	1.1708e−1	3.0490e−1	2.5299e−1	**5.7784e−2**
	(1.04e−1) −	(8.02e−2) −	(1.43e−1) −	(6.31e−2) −	(6.87e−2) −	(1.04e−1) −	(9.34e−2) −	(1.19e−1) −	**(1.29e−2)**
IDMPM4T3	1.5320e−1	6.7522e−1	1.8971e−1	1.1866e−1	8.9133e−2	1.3361e−1	3.8119e−1	3.2367e−1	**5.4516e−2**
	(1.72e−2) −	(1.14e−1) −	(1.06e−1) −	(1.38e−2) −	(4.00e−2) −	(1.21e−1) −	(1.24e−1) −	(1.64e−1) −	**(4.70e−3)**
IDMPM4T4	3.5198e−1	7.7223e−1	1.9025e−1	2.1309e−1	8.4755e−2	3.5695e−1	2.8889e−1	3.1973e−1	**5.0915e−2**
	(1.06e−1) −	(4.53e−2) −	(1.44e−1) −	(7.17e−2) −	(5.72e−2) −	(1.39e−1) −	(1.03e−1) −	(1.48e−1) −	**(1.64e−3)**
IDMPM2T1_e	3.3079e−2	3.5302e−2	3.4308e−2	3.3004e−2	3.9123e−3	3.4460e−2	1.2679e−2	**3.7824e−3**	4.1139e−3
	(3.61e−4) −	(3.83e−5) −	(2.12e−4) −	(3.15e−4) −	(2.11e−4) +	(1.37e−4) −	(6.84e−3) −	**(2.24e−4)** +	(1.98e−4)
IDMPM2T2_e	3.4054e−2	3.5332e−2	3.4407e−2	3.3726e−2	**4.0298e−3**	3.4721e−2	1.1967e−2	4.2154e−3	4.0448e−3
	(1.88e−4) −	(2.33e−5) −	(1.35e−4) −	(1.43e−4) −	**(1.49e−4)** =	(1.02e−4) −	(7.23e−3) −	(1.18e−4) −	(1.61e−4)
IDMPM2T3_e	8.6374e−2	2.6502e−1	2.5010e−2	3.1512e−2	6.7202e−3	2.6343e−2	1.2021e−2	6.3912e−3	**6.1187e−3**
	(9.90e−2) −	(1.76e−1) −	(1.19e−4) −	(6.00e−3) −	(1.82e−4) −	(2.66e−4) −	(5.07e−3) −	(1.40e−4) −	**(1.72e−4)**
IDMPM2T4_e	5.9807e−1	5.8955e−1	5.9912e−1	5.9827e−1	4.7350e−2	5.9920e−1	2.2019e−2	**1.4365e−2**	2.8260e−2
	(3.71e−4) −	(3.53e−2) −	(1.01e−4) −	(2.78e−4) −	(5.39e−2) −	(1.55e−4) −	(2.46e−3) +	**(8.30e−4)** +	(3.55e−2)
IDMPM3T1_e	1.5987e−1	3.1990e−1	1.3134e−1	1.3541e−1	4.7792e−2	1.3299e−1	4.4871e−2	**4.1330e−2**	4.2722e−2
	(1.28e−2) −	(1.93e−1) −	(6.51e−4) −	(3.75e−3) −	(1.06e−3) −	(5.81e−4) −	(1.34e−3) −	**(5.92e−4)** +	(6.68e−4)
IDMPM3T2_e	1.8731e−1	3.6971e−1	1.6477e−1	1.6949e−1	4.1092e−2	1.6549e−1	3.9476e−2	7.3351e−2	**3.7402e−2**
	(1.78e−2) −	(8.96e−2) −	(7.64e−4) −	(5.76e−3) −	(8.94e−4) −	(6.66e−4) −	(3.02e−3) −	(6.02e−4) −	**(6.36e−4)**
IDMPM3T3_e	1.6772e−1	2.8983e−1	1.6477e−1	1.6349e−1	4.3548e−2	1.6487e−1	5.9895e−2	7.2763e−2	**3.8609e−2**
	(3.12e−3) −	(1.09e−1) −	(7.39e−4) −	(1.05e−3) −	(6.05e−3) −	(6.92e−4) −	(9.78e−3) −	(5.20e−4) −	**(7.44e−4)**
IDMPM3T4_e	5.1708e−1	6.1406e−1	5.1000e−1	5.1329e−1	6.1064e−2	5.5526e−1	6.2282e−2	1.7087e−1	**5.6309e−2**
	(1.12e−2) −	(9.71e−2) −	(4.21e−4) −	(7.09e−3) −	(4.27e−3) −	(8.31e−2) −	(2.19e−3) −	(6.58e−4) −	**(1.14e−3)**
Polygon-2	1.8785e−1	1.9097e−1	1.6892e−1	1.6943e−1	2.9836e−3	1.5914e−1	**2.9358e−3**	1.7587e−1	2.9781e−3
	(7.39e−3) −	(2.72e−5) −	(2.31e−2) −	(1.73e−2) −	(1.05e−4) =	(2.55e−2) −	**(8.30e−5)** =	(1.83e−2) −	(1.20e−4)
Polygon-3	2.7514e−1	3.0017e−1	2.2602e−1	2.5670e−1	4.9881e−2	2.9086e−1	4.0880e−2	2.4738e−1	**4.0178e−2**
	(1.74e−2) −	(5.73e−3) −	(2.03e−2) −	(2.06e−2) −	(1.68e−2) −	(1.12e−2) −	(6.91e−4) −	(2.17e−2) −	**(7.85e−4)**
Polygon-4	2.6772e−1	3.1514e−1	2.0931e−1	2.6390e−1	5.1692e−2	3.0042e−1	4.7787e−2	2.3392e−1	**4.7638e−2**
	(2.34e−2) −	(5.29e−3) −	(2.22e−2) −	(2.92e−2) −	(8.68e−3) −	(4.73e−3) −	(9.22e−4) =	(3.12e−2) −	**(1.16e−3)**
Polygon-5	2.9694e−1	3.9989e−1	2.4736e−1	2.9630e−1	7.0214e−2	3.6510e−1	**6.7995e−2**	2.6623e−1	6.8264e−2
	(3.24e−2) −	(6.54e−3) −	(2.62e−2) −	(2.87e−2) −	(2.13e−3) −	(2.69e−3) −	**(7.66e−4)** =	(3.63e−2) −	(1.05e−3)
Polygon-6	3.1707e−1	4.1693e−1	2.3402e−1	2.9649e−1	7.5182e−2	3.7412e−1	**7.2168e−2**	2.7218e−1	7.2729e−2
	(2.96e−2) −	(1.65e−3) −	(1.93e−2) −	(3.03e−2) −	(1.87e−3) −	(1.70e−2) −	**(8.62e−4)** =	(3.19e−2) −	(1.08e−3)
Polygon-8	3.4102e−1	5.1208e−1	2.3834e−1	3.1562e−1	8.7881e−2	4.3374e−1	**8.4643e−2**	2.8813e−1	8.5790e−2
	(3.69e−2) −	(2.20e−2) −	(2.04e−2) −	(4.18e−2) −	(1.88e−3) −	(2.22e−2) −	**(1.13e−3)** +	(3.27e−2) −	(1.27e−3)
Polygon-10	3.5806e−1	5.8567e−1	2.4782e−1	3.3270e−1	1.0143e−1	4.8399e−1	**9.5278e−2**	3.0365e−1	9.7001e−2
	(4.72e−2) −	(1.62e−2) −	(2.23e−2) −	(4.39e−2) −	(7.05e−3) −	(3.07e−2) −	**(9.81e−4)** +	(3.23e−2) −	(1.43e−3)
Polygon-12	3.6712e−1	6.1588e−1	2.5791e−1	3.5337e−1	1.0862e−1	4.9213e−1	**1.0467e−1**	3.1988e−1	1.0619e−1
	(4.35e−2) −	(1.12e−2) −	(2.11e−2) −	(4.71e−2) −	(3.21e−3) −	(6.58e−2) −	**(1.02e−3)** +	(4.32e−2) −	(1.27e−3)
+/−/=	0/28/0	0/28/0	0/28/0	0/28/0	1/24/3	0/27/1	4/20/4	7/21/0	

Fig. 10 shows the equivalent PSs obtained by different algorithms on Polygon-12. Polygon-12 has three local PSs and one global PS, and the manifolds of different PSs are different, which challenges its ability to balance convergence and diversity in the decision space. It can be seen from Fig. 10 that only CDP-BCD, RHEA, and MMOEA/DC can search all local PSs. Since the distance between the local PF and the global PF in Polygon-12 is relatively large, and HREA will directly delete the solutions on the local PF when performing hierarchical sorting, the performance of HREA will be worse when the distance between the local PF and the global PF is relatively large. The existence of the convergence-first mechanism in MO_Ring_PSO_SCD, TriMOEA-TA&R, MMODE_ICD, and MMEA-WI makes them automatically delete the local PS, so only the global PS can be searched.

5.2. Comparative results on the MMF

Table 4 is the IGDX indicator result obtained by the comparison algorithm on the MMF, respectively. CDP-BCD has the best IGDX value on four benchmark problems, with only global PS. HREA obtained the five best IGDX values among 19 benchmark problems, of which three out of five benchmark problems have the local PS. RHEA has the best IGDX value on five benchmark problems, two of which have the local PS. Compared with HREA, CDP-BCD's IGDX result is slightly better than HREA, but the gap between HREA and CDP-BCD on IGDX is small on most benchmark problems. But on MMFs with the local PS, the IGDX result of CDP-BCD is slightly worse than HREA. The main reason is that when CDP-BCD uses DSS to select solutions, it will prefer solutions

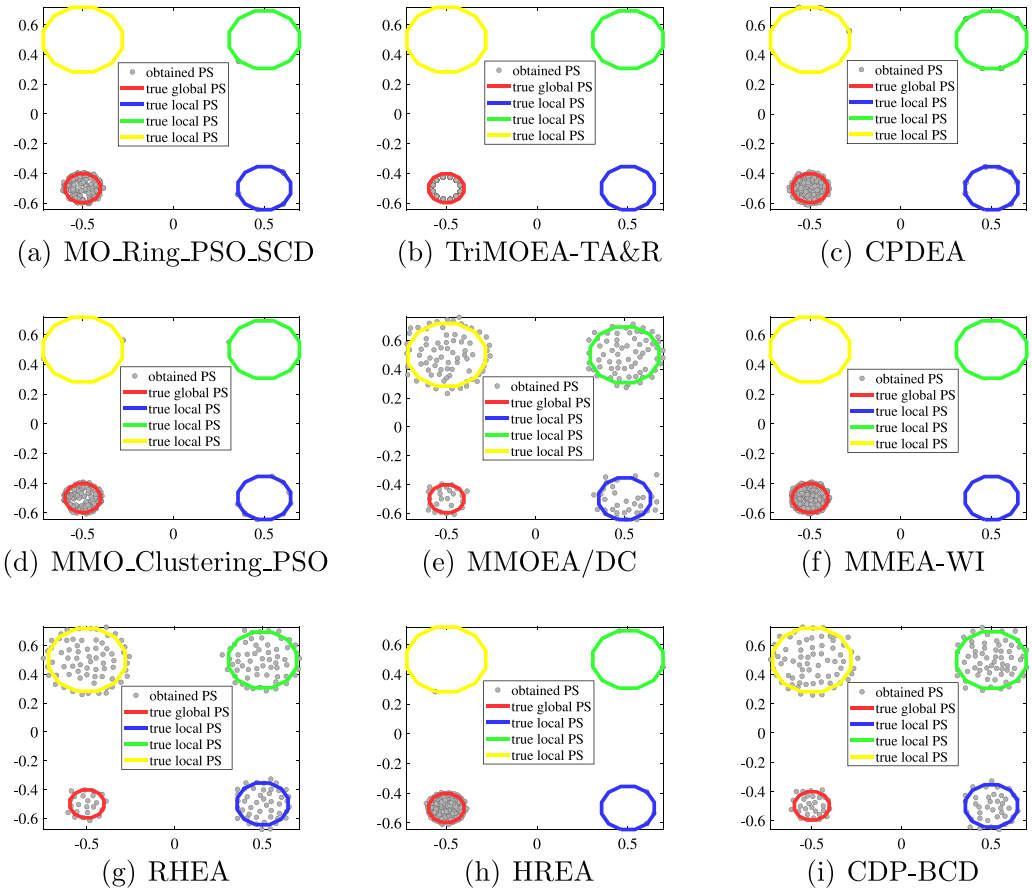


Fig. 10. Distribution of the equivalent PSs obtained by different algorithms on Polygon-12.

with slightly less convergence to a certain extent because solutions with slightly less convergence will have better diversity. RHEA is better than CDP-BCD's IGDX result on the MMF9, MMF10, and MMF11 benchmark problems, mainly because the size of the objective space of MMF9, MMF10, and MMF11 is larger than that of the decision space. Using DSS will select solutions with good diversity in the objective space, which leads to poor performance of CDP-BCD in the decision space. Finally, MMOEA/DC is also very competitive on the IGDX. In the later stage of evolution, the clustering operation of MMOEA/DC in the decision space enables each subclass to locate a PS (global or local PS). Moreover, in the environmental selection, the dominance relationship will only be used in each subclass, which can reduce the impact of the global PS on searching for a local PS so that MMOEA/DC can search all global and local PSs.

Table 5 is the IGDM indicator result on the MMF. As seen from Table 5, CDP-BCD performed the best and has the five best IGDM values. CPDEA has the best IGDX value on six benchmark problems. RHEA and MMOEA/DC have the four best IGDM values, respectively. It is worth noting that on MMF9, MMF10, MMF11, and MMF13, the IGDM values of MMOEA/DC and RHEA are better than CDP-BCD. This is because the objective space of these problems is larger than the decision space. When DSS balances the diversity between the decision and objective spaces, the solutions with good diversity in the objective space will be selected, resulting in worse convergence and diversity. While HREA only pays attention to the diversity in the decision space, since the diversity between the decision space and the objective space in the MMF benchmark problem is very imbalanced, HREA has poor IGDM values on MMFs. Furthermore, CPDEA uses the double k-nearest neighbor method to evaluate diversity quality both in the objective and decision spaces of the population so that CPDEA can balance the diversity between the decision and objective spaces, so CPDEA also has better IGDM values on the MMFs with only the global PS, especially on MMF1, MMF1_e, MMF5, MMF14, and MMF14_a.

As can be seen from Tables 4 and 5, the IGDX results of CDP-BCD on MMF4 and MMF6 are worse than other algorithms, but the IGDM results on MMF4 and MMF6 are better than other algorithms. HREA only focuses on diversity in the decision space, while CDP-BCD has a mechanism to balance the diversity between the decision and objective spaces. Therefore, on MMF4, MMF12, MMF13, and MMF15, the IGDX results of CDP-BCD are worse than HREA, and the IGDM results of CDP-BCD are better than HREA. In MMF5 and MMF6, the manifold of PS is more complex than that of PF, and there is coverage between PSs in every dimension. When solving MMF5 and MMF6, CDP-BCD uses DSS to balance the diversity between the decision and objective spaces. The diversity of the objective space is more important than the diversity of the decision space. Therefore, on MMF5 and MMF6, the IGDX results of CDP-BCD are worse than RHEA, and the IGDM results of CDP-BCD are better than RHEA. Although DSS does not require parameters to maintain diversity, when encountering the PS or PF with complex manifolds, it will be more difficult for DSS to maintain the diversity among solutions within the niche on the MMF. Therefore, the overall performance of CDP-BCD on the MMF will be worse than that of the IDMP. Since Tables 4 and 5 are not enough to show the advantages of CDP-BCD on the MMF, we give the average rank of IGDX and IGDM on the MMF obtained by comparative algorithms as shown in Table 6. As seen from Table 6, CDP-BCD has the most minor ranking on both IGDX and IGDM, so CDP-BCD also has an advantage on the MMF.

Fig. 11 shows the distribution of the equivalent PSs obtained by different algorithms on MMF15_a. It can be seen from Fig. 11 that CDP-BCD, HREA, RHEA, and MMOEA/DC can search for a local PS. CDP-BCD, HREA, RHEA, and MMOEA/DC all find the solution with the best convergence from the local range, so they will not be affected

Table 4
IGDX results obtained by compared algorithms on the MMF.

Problem	MO_PSO_Ring_SCD	TriMOEA-TA&R	CPDEA	MMO_Clust_ering_PSO	MMOEA/DC	MMEA-WI	RHEA	HREA	CDP-BCD
MMF1	6.2695e−2 (5.99e−3) −	5.9712e−2 (4.70e−3) −	3.3266e−2 (6.82e−4) −	4.1326e−2 (2.09e−3) −	4.4021e−2 (1.24e−3) −	4.0839e−2 (2.39e−3) −	3.3341e−2 (9.49e−4) −	3.3857e−2 (1.18e−3) −	**3.2772e−2 (8.36e−4)**
MMF1_e	2.4184e−1 (4.05e−2) =	2.8829e−1 (1.43e−1) −	2.2167e−1 (3.90e−2) =	2.2736e−1 (4.54e−2) =	**2.1636e−1 (4.57e−2)** =	2.9829e−1 (4.69e−2) −	2.6877e−1 (3.90e−2) −	2.3707e−1 (5.96e−2) =	2.2644e−1 (4.74e−2)
MMF1_z	4.7543e−2 (7.90e−3) −	6.1801e−2 (1.70e−2) −	2.5244e−2 (8.27e−4) −	3.1473e−2 (2.31e−3) −	2.9693e−2 (1.65e−3) −	3.0918e−2 (1.85e−3) −	2.4548e−2 (1.09e−3) −	2.3944e−2 (1.05e−3) −	**2.2677e−2 (6.37e−4)**
MMF2	8.2927e−2 (4.42e−2) −	6.7276e−2 (2.85e−2) −	1.6360e−2 (3.00e−3) +	2.4798e−2 (9.22e−3) −	**1.2376e−2 (2.81e−3)** +	2.1587e−2 (8.86e−3) =	3.2709e−2 (9.15e−3) −	2.4698e−2 (7.94e−3) −	2.2042e−2 (1.24e−2)
MMF3	5.1619e−2 (2.28e−2) −	5.1507e−2 (2.23e−2) −	1.3700e−2 (1.52e−3) +	1.7901e−2 (5.12e−3) =	**1.1137e−2 (1.94e−3)** +	1.7238e−2 (5.37e−3) =	2.0470e−2 (4.05e−3) =	1.8423e−2 (3.27e−3) =	1.9624e−2 (1.07e−2)
MMF4	4.6834e−2 (8.53e−3) −	1.3347e−1 (1.75e−1) −	1.8952e−2 (6.04e−4) =	2.4619e−2 (1.50e−3) −	2.4813e−2 (3.74e−3) −	2.3828e−2 (1.10e−3) −	2.1736e−2 (6.26e−4) −	**1.8327e−2 (5.31e−4)** +	1.9131e−2 (6.28e−4)
MMF5	1.1156e−1 (1.07e−2) −	9.7275e−2 (9.89e−3) −	6.0154e−2 (1.12e−3) −	7.3209e−2 (2.82e−3) −	7.6677e−2 (1.82e−3) −	7.0097e−2 (1.82e−3) −	**5.7391e−2 (1.30e−3)** +	5.7929e−2 (2.18e−3) =	5.8339e−2 (1.43e−3)
MMF6	9.5513e−2 (8.17e−3) −	8.0164e−2 (5.20e−3) −	5.5224e−2 (9.81e−4) −	6.5851e−2 (2.62e−3) −	6.7298e−2 (2.06e−3) −	6.2635e−2 (2.40e−3) −	**5.2487e−2 (9.03e−4)** +	5.3590e−2 (9.76e−4) =	5.3677e−2 (9.46e−4)
MMF7	4.1731e−2 (8.61e−3) −	4.2749e−2 (2.62e−2) −	1.8863e−2 (6.23e−4) =	2.6368e−2 (2.32e−3) −	2.6702e−2 (3.31e−3) −	2.3517e−2 (1.73e−3) −	**1.8609e−2 (5.46e−4)** +	1.9058e−2 (7.22e−4) =	1.8986e−2 (6.19e−4)
MMF8	1.3563e−1 (5.89e−2) −	4.9720e−1 (1.19e−1) −	4.1512e−2 (2.13e−3) −	5.1206e−2 (5.60e−3) −	6.1112e−2 (1.54e−2) −	5.9252e−2 (1.08e−2) −	4.3149e−2 (1.73e−3) −	4.2463e−2 (3.39e−3) −	**3.7326e−2 (2.40e−3)**
MMF9	1.4470e−2 (4.78e−3) −	**3.2955e−3 (1.08e−4)** +	6.3972e−3 (2.75e−4) −	7.2626e−3 (6.41e−4) −	6.6155e−3 (2.37e−4) −	6.2637e−3 (2.99e−4) =	6.0376e−3 (1.59e−4) +	9.2875e−3 (5.14e−4) −	6.2344e−3 (2.68e−4)
MMF10	1.7128e−1 (1.09e−2) −	2.0146e−1 (3.93e−5) −	2.0091e−1 (3.31e−5) −	1.6565e−1 (3.52e−3) −	9.8904e−3 (6.49e−4) −	1.9766e−1 (1.06e−2) −	**7.3615e−3 (3.66e−4)** +	8.0536e−3 (5.40e−4) +	9.2564e−3 (4.45e−4)
MMF11	2.3374e−1 (2.09e−2) −	2.5244e−1 (5.83e−5) −	2.4836e−1 (2.91e−3) −	2.1709e−1 (2.60e−2) −	7.4811e−3 (2.67e−4) +	2.4896e−1 (2.15e−4) −	**6.7202e−3 (2.27e−4)** +	1.0879e−2 (2.41e−3) −	8.5719e−3 (3.33e−4)
MMF12	2.2205e−1 (3.42e−2) −	2.4803e−1 (2.20e−4) −	2.4499e−1 (2.56e−4) −	1.9731e−1 (4.49e−2) −	3.1974e−3 (1.55e−4) =	2.4445e−1 (3.52e−4) −	2.8852e−3 (1.24e−4) +	**2.7981e−3 (1.20e−4)** +	3.1692e−3 (1.72e−4)
MMF13	2.4859e−1 (1.24e−2) −	2.6882e−1 (6.89e−3) −	2.5181e−1 (4.45e−4) −	2.3193e−1 (1.34e−2) −	9.2684e−2 (3.59e−2) −	2.5148e−1 (5.62e−4) −	8.8760e−2 (2.30e−3) −	**5.0624e−2 (2.94e−3)** +	5.7242e−2 (1.85e−3)
MMF14	6.2866e−2 (2.90e−3) −	**3.6951e−2 (4.69e−4)** +	3.8806e−2 (4.04e−4) +	5.4117e−2 (4.44e−3) −	5.1444e−2 (1.24e−3) −	4.2704e−2 (6.43e−4) −	4.3881e−2 (5.67e−4) −	3.9719e−2 (9.56e−4) +	4.0757e−2 (5.45e−4)
MMF14_a	6.9092e−2 (3.15e−3) −	5.2599e−2 (1.09e−3) −	5.0390e−2 (5.01e−4) −	5.9753e−2 (1.76e−3) −	7.6184e−2 (2.72e−3) −	5.7038e−2 (1.65e−3) −	5.1283e−2 (4.96e−4) −	**4.7487e−2 (4.78e−4)** +	4.9930e−2 (6.36e−4)
MMF15	1.9874e−1 (1.86e−2) −	2.7130e−1 (2.90e−4) −	2.2568e−1 (1.69e−2) −	1.7079e−1 (2.01e−2) −	5.3601e−2 (1.70e−3) −	2.5811e−1 (9.34e−4) −	4.6019e−2 (5.41e−4) −	**4.5122e−2 (1.83e−3)** =	4.5423e−2 (6.39e−4)
MMF15_a	1.9012e−1 (1.01e−2) −	2.1850e−1 (9.41e−3) −	2.0226e−1 (6.72e−3) −	1.6780e−1 (9.77e−3) −	1.1104e−1 (3.82e−2) −	2.0893e−1 (2.59e−3) −	5.7368e−2 (1.39e−3) −	5.6224e−2 (2.76e−3) −	**5.3214e−2 (8.21e−4)**
+/−/=	0/18/1	2/17/0	3/13/3	0/17/2	3/14/2	0/16/3	7/11/1	6/7/6	

by the global PS when searching for the local PS. To a certain extent, the drawbacks brought by the convergence-first mechanism are avoided (as shown in Fig. 2). In comparison, the convergence-first mechanism in MO_Ring_PSO_SCD, TriMOEA-TA&R, CPDEA, and MMEA-WI makes them lose the local PS. Although MMO_Clustering_PSO is designed to solve MMOPL, the convergence-first mechanism still makes the diversity of the local PSs searched poor.

5.3. Discussion on the weak interaction coevolutionary mechanism

Considering that excessive interaction between populations in the traditional weak interaction mechanism will lead to poor convergence of the population and even the loss of PSs that are difficult to search, we designed a new weak interaction mechanism. The new weak interaction mechanism can enhance the convergence quality of the population and retain PSs that are more difficult to search. Therefore, to verify the advantages of the new weak interaction mechanism, we compared it with the traditional weak interaction mechanism on the MMOP-ICD and Polygon-based MMOPL.

The CDP-BCD-weak in Fig. 12 and Table 7 refers to the CDP-BCD combined with the traditional weak interaction mechanism. It can be seen from Fig. 12 that the newly proposed weak interaction mechanism performs better on the MMOP-ICD and Polygon-based MMOPL. In the traditional weak interaction mechanism, the auxiliary population and the main population interact at the beginning of evolution, so the exploration and development between PSs influence each other, resulting in an extreme imbalance between convergence and diversity in the decision space. In the new weak interaction mechanism, since the interaction between the main population and the auxiliary population is carried out after the auxiliary population locates the position of the PS, the exploration and development between PSs do not affect each other. As can be seen from Table 7, the performance of CDP-BCD-weak on IDMPM2s is very poor and even worse on IDMPM4s due to the loss of PS, so the new weak interaction mechanism can bring better convergence to the population.

5.4. Discussion on the strength local convergence quality

When updating the auxiliary population, the role of SLCQ is to help the algorithm locate the positions of the true PSs. Exploring the location of the PS in the early stage helps develop the PS when updating the main population in the later stage. In addition, when the dimension becomes larger and the difference in search difficulty of different PSs becomes more significant, the PS with higher search difficulty is ignored. Based on these considerations, we add the Euclidean distance to the solution dominating X when calculating the SLCQ value of X. Adding Euclidean distance helps the algorithm keep the solutions around the imbalanced PS with greater search difficulty.

Since the calculation of SLCQ is similar to the local convergence quality in HREA, we replace SLCQ in CDP-BCD with local convergence quality in HREA (v2 in Fig. 13) and compare with the standard CDP-BCD (v1 in Fig. 13) on several benchmark problems. IGDM and IGDX results are the mean of 30 independent runs of v1 and v2 on each benchmark problem. It can be seen from the figure that the IGDX and IGDM obtained by standard CDP-BCD in IDMPM2T4_e are worse than v2. Because the number of PSs contained in IDMPM2T4_e is too

Table 5
IGDM results obtained by compared algorithms on the MMF.

Problem	MO_PSO_Ring_SCD	TriMOEA-TA&R	CPDEA	MMO_Clustering_PSO	MMOEA/DC	MMEA-WI	RHEA	HREA	CDP-BCD
MMF1	1.1613e−2 (1.72e−3) −	8.9576e−3 (1.10e−3) −	**5.4621e−3 (1.97e−4)** +	7.2322e−3 (6.76e−4) −	7.2736e−3 (3.92e−4) −	7.4536e−3 (4.62e−4) −	6.2626e−3 (1.99e−4) −	6.8603e−3 (3.26e−4) −	6.0529e−3 (2.46e−4)
MMF1_e	5.9828e−2 (1.41e−2) −	1.0569e−1 (7.48e−2) −	**3.6259e−2 (2.67e−3)** +	4.1504e−2 (5.54e−3) =	6.0490e−2 (2.39e−2) −	6.8698e−2 (1.49e−2) −	5.1331e−2 (7.06e−3) −	4.0776e−2 (4.47e−3) =	4.2579e−2 (8.06e−3)
MMF1_z	1.3326e−2 (2.53e−3) −	1.2183e−2 (4.62e−3) −	7.2579e−3 (4.53e−4) −	7.4264e−3 (1.03e−3) −	7.0042e−3 (7.49e−4) −	8.7206e−3 (7.71e−4) −	6.9208e−3 (3.13e−4) −	7.1224e−3 (4.60e−4) −	**5.4253e−3 (2.20e−4)**
MMF2	8.6995e−2 (4.49e−2) −	6.7149e−2 (2.85e−2) −	1.6254e−2 (2.96e−3) +	2.6508e−2 (9.53e−3) −	**1.3457e−2 (2.57e−3)** +	2.1587e−2 (8.79e−3) =	3.4371e−2 (8.96e−3) −	2.4597e−2 (7.90e−3) =	2.2134e−2 (1.23e−2)
MMF3	5.7827e−2 (3.09e−2) −	5.6722e−2 (3.25e−2) −	1.3640e−2 (1.50e−3) +	1.9014e−2 (5.07e−3) =	**1.2812e−2 (1.81e−3)** +	1.7359e−2 (5.43e−3) =	2.1700e−2 (4.03e−3) −	1.8376e−2 (3.17e−3) =	1.9618e−2 (1.07e−2)
MMF4	3.2331e−2 (8.72e−3) −	1.2839e−1 (1.87e−1) −	1.0721e−2 (3.04e−4) −	1.6010e−2 (1.48e−3) −	1.2962e−2 (1.60e−3) −	1.3732e−2 (6.62e−4) −	1.1196e−2 (2.75e−4) −	1.0834e−2 (3.23e−4) −	**9.9920e−3 (2.28e−4)**
MMF5	2.9209e−2 (8.20e−3) −	1.8706e−2 (3.08e−3) −	**1.0742e−2 (5.60e−4)** +	1.4254e−2 (5.94e−4) −	1.4511e−2 (6.08e−4) −	1.4029e−2 (7.90e−4) −	1.1584e−2 (5.40e−4) =	1.2331e−2 (7.97e−4) −	1.1476e−2 (4.15e−4)
MMF6	3.3520e−2 (9.41e−3) −	1.7775e−2 (4.91e−3) −	1.1791e−2 (4.13e−4) −	1.5303e−2 (1.43e−3) −	1.6082e−2 (9.21e−4) −	1.4329e−2 (7.11e−4) −	1.2132e−2 (4.13e−4) −	1.2889e−2 (7.43e−4) −	**1.1406e−2 (4.23e−4)**
MMF7	1.2443e−2 (4.41e−3) −	7.2622e−3 (1.30e−3) −	**5.8291e−3 (1.74e−4)** =	8.5994e−3 (8.16e−4) −	7.2585e−3 (8.48e−4) −	8.0083e−3 (4.57e−4) −	6.1847e−3 (2.44e−4) −	7.1865e−3 (4.31e−4) −	5.8946e−3 (1.33e−4)
MMF8	1.0380e−1 (3.46e−2) −	2.7212e−1 (5.81e−2) −	2.6209e−2 (1.55e−3) −	3.4096e−2 (3.56e−3) −	3.5906e−2 (8.75e−3) −	3.4805e−2 (5.58e−3) −	2.2589e−2 (8.35e−4) −	2.4695e−2 (1.63e−3) −	**2.2032e−2 (1.85e−3)**
MMF9	1.6312e−2 (5.24e−3) −	9.3974e−3 (6.93e−4) −	5.4108e−3 (1.72e−4) +	8.3637e−3 (1.22e−3) −	5.2154e−3 (1.69e−4) +	6.1471e−3 (3.19e−4) −	**4.8075e−3 (1.23e−4)** +	1.0662e−2 (8.29e−4) −	5.9159e−3 (2.88e−4)
MMF10	4.3781e−2 (4.08e−3) −	4.2874e−2 (2.56e−4) −	4.0806e−2 (5.13e−5) −	4.3436e−2 (3.35e−3) −	**3.9917e−3 (8.46e−5)** +	4.0664e−2 (1.07e−3) −	4.1235e−3 (1.13e−4) +	5.6333e−3 (4.03e−4) +	6.2708e−3 (3.17e−4)
MMF11	2.1190e−2 (4.81e−4) −	2.6445e−2 (6.07e−4) −	2.1953e−2 (5.89e−5) −	2.1327e−2 (6.27e−4) −	4.3932e−3 (1.05e−4) +	2.1960e−2 (2.65e−4) −	**4.1083e−3 (7.96e−5)** +	8.7907e−3 (2.12e−3) −	6.2113e−3 (1.87e−4)
MMF12	4.8561e−2 (6.83e−3) −	5.5413e−2 (3.18e−4) −	5.3723e−2 (1.73e−4) −	4.4560e−2 (6.92e−3) −	2.6793e−3 (8.69e−5) +	5.4004e−2 (2.90e−4) −	**2.4307e−3 (4.91e−5)** +	4.0313e−3 (2.26e−4) −	2.9393e−3 (7.99e−5)
MMF13	2.0304e−2 (1.27e−3) −	2.9105e−2 (5.22e−4) −	2.3661e−2 (1.42e−4) −	1.9456e−2 (1.01e−3) −	**4.3886e−3 (3.26e−3)** +	2.3785e−2 (2.45e−4) −	4.5143e−3 (5.11e−4) +	6.1869e−3 (1.27e−3) =	6.6112e−3 (7.60e−4)
MMF14	6.6437e−2 (3.51e−3) −	4.4288e−2 (5.14e−4) −	**4.1546e−2 (4.67e−4)** +	5.6987e−2 (2.41e−3) −	4.8942e−2 (7.72e−4) −	4.8177e−2 (8.75e−4) −	4.4816e−2 (5.51e−4) −	4.4795e−2 (1.16e−3) −	4.2791e−2 (5.56e−4)
MMF14_a	6.1514e−2 (2.76e−3) −	4.7395e−2 (7.11e−4) −	**4.3641e−2 (4.45e−4)** +	5.4546e−2 (1.65e−3) −	5.0491e−2 (1.31e−3) −	5.1517e−2 (9.51e−4) −	4.5533e−2 (5.46e−4) −	4.5754e−2 (5.37e−4) −	4.4891e−2 (3.79e−4)
MMF15	7.6133e−2 (1.51e−3) −	8.9372e−2 (2.94e−4) −	7.3392e−2 (1.17e−3) −	7.5155e−2 (1.30e−3) −	4.3170e−2 (1.07e−3) −	7.9238e−2 (9.99e−4) −	**3.9966e−2 (3.60e−4)** +	4.4083e−2 (2.12e−3) −	4.1346e−2 (4.90e−4)
MMF15_a	7.6103e−2 (1.95e−3) −	8.2637e−2 (8.42e−3) −	7.2543e−2 (8.52e−4) −	7.5385e−2 (1.53e−3) −	6.0527e−2 (6.44e−3) −	7.6731e−2 (1.26e−3) −	4.5421e−2 (8.72e−4) =	5.0909e−2 (2.81e−3) −	**4.5245e−2 (7.39e−4)**
+/−/=	0/19/0	0/19/0	8/10/1	0/17/2	7/12/0	0/17/2	6/11/2	1/14/4	

Table 6
The average rank of IGDX and IGDM on the MMF obtained by comparative algorithms.

Algorithm	MO_Ring_PSO_SCD	TriMOEA-TA&R	CPDEA	MMO_Clustering_PSO	MMOEA/DC	MMEA-WI	RHEA	HREA	CDP-BCD
IGDX	7.6316	7.6316	4.2632	5.6842	4.9474	5.8947	3.3158	2.9474	2.6842
IGDM	8	7.7895	3.3158	6.0526	4.0526	6.3158	3	3.8947	2.5789

Table 7
The IGDX and IGDM results of solutions obtained by CDP-BCD and its variant on IDMPM2Ts and IDMPM4Ts respectively over 30 runs.

Problem	CDP-BCD-weak	CDP-BCD	CDP-BCD-weak	CDP-BCD
	IGDX		IGDM	
IDMPM2T1	6.0395e−4 (6.22e−5) −	**5.8525e−4 (2.43e−5)**	4.3915e−3 (5.04e−4) −	**4.2537e−3 (2.72e−4)**
IDMPM2T2	7.0942e−4 (4.35e−5) −	**7.0488e−4 (3.89e−5)**	4.0333e−3 (7.13e−5) =	**4.0206e−3 (7.53e−5)**
IDMPM2T3	1.1704e−3 (1.85e−4) −	**1.0824e−3 (7.76e−5)**	5.6121e−3 (1.10e−3) −	**5.0599e−3 (5.03e−4)**
IDMPM2T4	6.0160e−4 (5.41e−5) −	**5.8959e−4 (4.80e−5)**	4.2531e−3 (4.55e−4) −	**4.1407e−3 (3.89e−4)**
IDMPM4T1	**6.7796e−3 (1.12e−4)** +	7.2898e−3 (4.32e−4)	**5.1792e−2 (1.59e−3)** +	5.7736e−2 (4.63e−3)
IDMPM4T2	9.5060e−2 (1.27e−1) −	**8.3100e−3 (2.78e−3)**	1.3033e−1 (1.10e−1) −	**6.0412e−2 (1.97e−2)**
IDMPM4T3	7.2119e−2 (1.15e−1) −	**8.5043e−3 (6.28e−4)**	1.1530e−1 (1.01e−1) −	**5.6287e−2 (5.68e−3)**
IDMPM4T4	2.3409e−1 (1.74e−1) −	**7.2375e−3 (3.77e−4)**	2.4930e−1 (1.54e−1) −	**5.0994e−2 (4.00e−3)**
+/−/=	1/7/0		1/6/1	

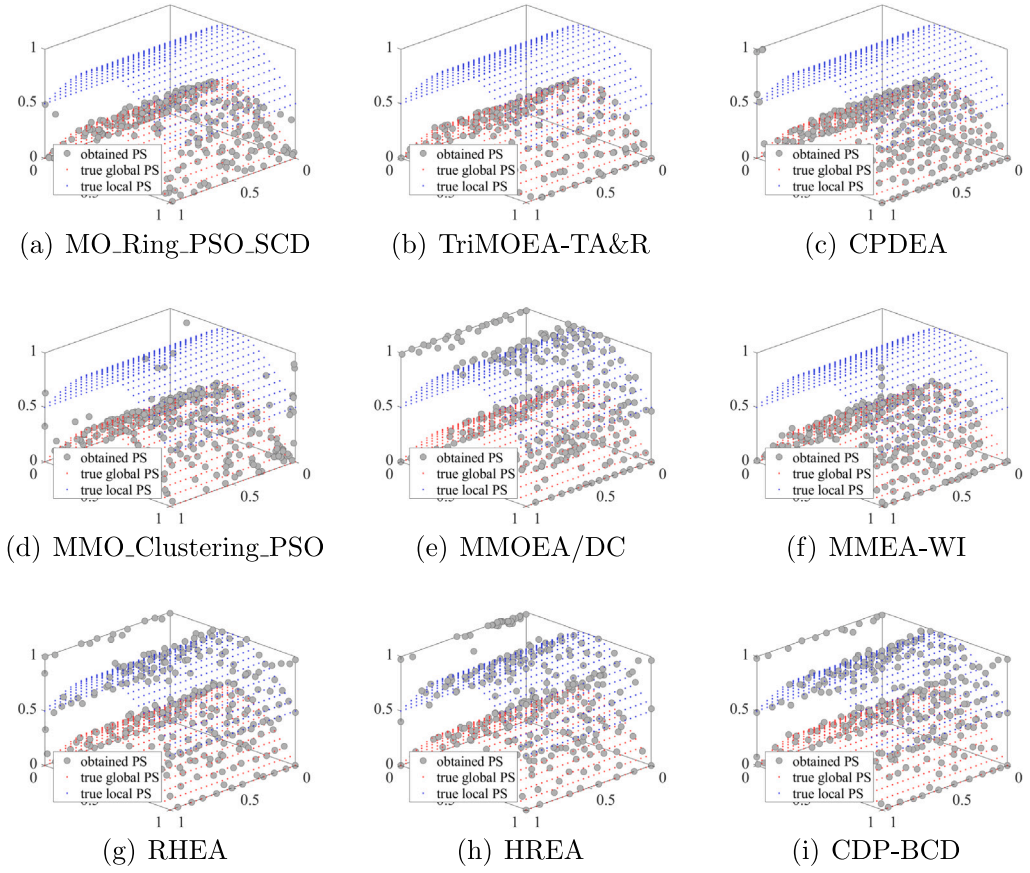


Fig. 11. Distribution of the equivalent PSs obtained by different algorithms on MMF15_a.

	Firedman's ranking (IGDX)	Firedman's ranking (IGDM)
CDP-BCD-weak	1.7321	1.7143
CDP-BCD	1.2679	1.2857

Fig. 12. The average rank of solutions obtained by CDP-BCD and its variant on the MMOP-ICD and Polygon-based MMOPL respectively over 30 runs.

large, and the local radius in SLCQ in the early stage of the algorithm is larger than the local radius in LCQ in HREA, this will cause CDP-BCD to search for a local PS in the early stage of the algorithm to be affected by other global or local PSs so that CDP-BCD will lose some local PSs. Nevertheless, on most benchmark problems, standard CDP-BCD obtained better IGDX and IGDM values than v2. Therefore, the proposed local convergence quality is valid.

To demonstrate the ability of SLCQ to locate the true PSs when updating the auxiliary population, Fig. 14 records the results of the auxiliary population on the IDMPM4 benchmark problems (MMOP-ICD) using different local convergence quality assessment schemes. The dimensions of the decision space and the objective space of the IDMPM4 benchmark problem are both 4, and the difference in the search difficulty between different PSs is the smallest in the IDMPM4T1 benchmark problem, while the difference in the search difficulty between different PSs is the largest in the IDMPM4T4 benchmark problem. Lconv1 means the SLCQ in CDP-BCD is not replaced, Lconv2 means the SLCQ in CDP-BCD is replaced by the local convergence quality in HREA, and Lconv3 means the SLCQ value in CDP-BCD is replaced by the local convergence quality in RHEA. Since the visualization results in 4D space are not intuitive, we record the first and second dimensions of IDMPM4 benchmark problems. The experimental population size shown in Fig. 14 is set to 400, and the maximum number of evaluations is 40,000.

It can be seen from Fig. 14 that Lconv3 can only locate the position of PSs with less search difficulty, and imbalanced PSs with greater search difficulty are deleted. And Lconv2 can locate the position of equivalent PSs only when the gap of search difficulty between equivalent PS is small, such as IDMPM4T1. When the search difficulty gap between PSs is large, the ability of Lconv2 to locate the equivalent PS becomes worse. For example, when encountering IDMPM4T2, IDMPM4T3, and IDMPM4T4, Lconv2 cannot locate the position of the equivalent PSs. Lconv1 can help the auxiliary population locate all the PSs. Even when the search difficulty gap between PSs becomes larger, Lconv1 still can locate equivalent PSs. Therefore, when the dimensionality of MMOP-ICD becomes larger and the difference in search difficulty between equivalent PSs becomes larger, SLCQ can well locate all equivalent PSs.

5.5. Discussion on the niche-based truncation strategy

The role of NBT is to remove those solutions that do not contribute significantly to convergence. Local areas of these removed solutions

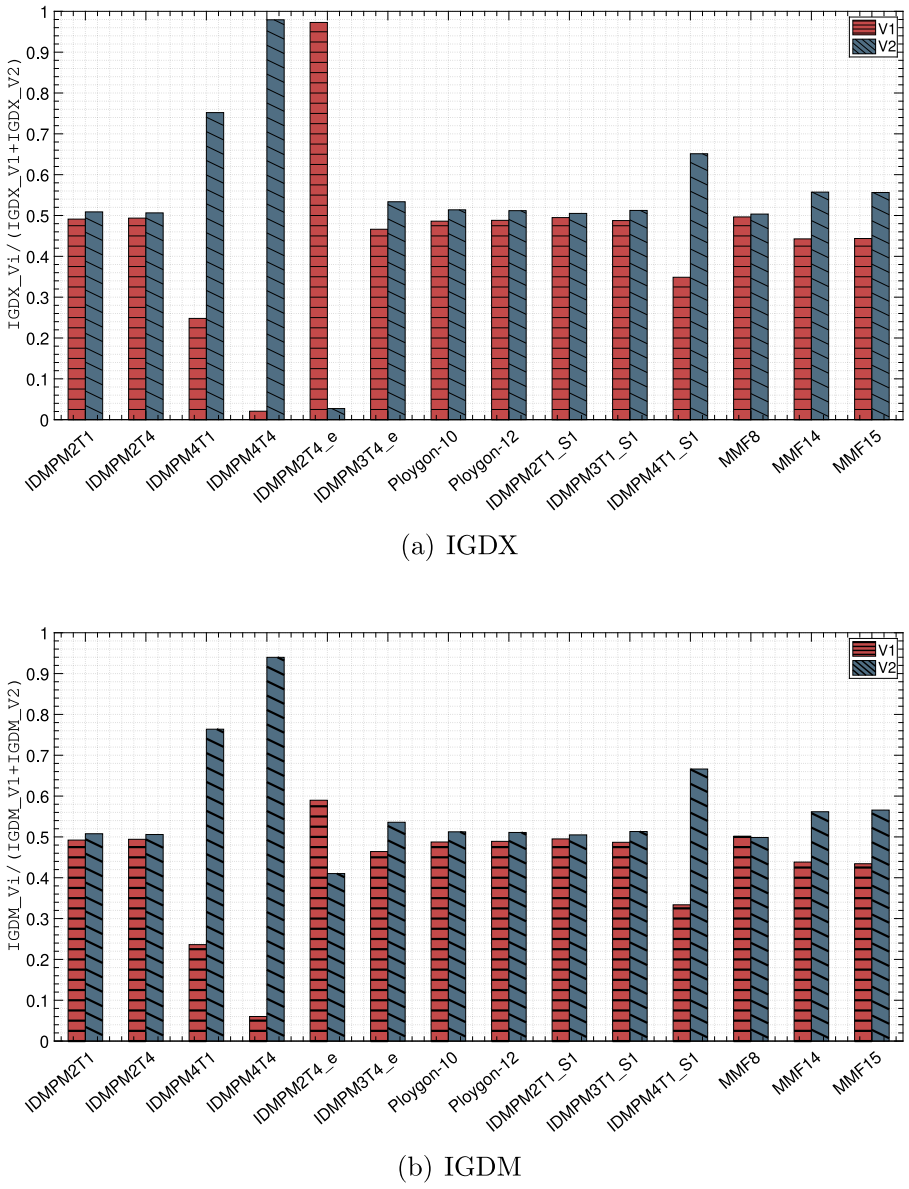


Fig. 13. The performance of CDP-BCD with various local convergence qualities.

have other solutions with better convergence for exploration and development. If these removed solutions are kept, search resources will be largely wasted. Therefore, NBT can improve population convergence to a certain extent. To verify the effectiveness of NBT, we removed NBT from CDP-BCD and compared it with standard CDP-BCD on several benchmark problems. The ablation experiments of NBT adopt IGDX and IGDM performance indicators, and all experimental results are obtained by running the algorithm 30 times independently on each benchmark problem. Table 8 shows the ablation experiment results of NBT on the MMOP-ICD and Polygon-based MMOPL. It can be seen from Table 8 that the CDP-BCD performed worse when the NBT is removed. Since IGDX can measure the convergence and diversity of the population in the decision space, and IGDM can simultaneously calculate the diversity and convergence of the population both in the decision space and the objective space, NBT makes a more significant contribution to improving the convergence and diversity of the population.

5.6. Discussion on the distance-based subset selection scheme

When the diversity between the decision space and the objective space is imbalanced, the algorithm will have good diversity in one space and poor diversity in the other space. Table 9 shows the results of DSS used in different spaces on several benchmark problems. In the DSS of CDP-BCD-Obj, $d(x_i, x_j) = d_{obj}(x_i, x_j)$. In the DSS of CDP-BCD-Dec, $d(x_i, x_j) = d_{dec}(x_i, x_j)$. The IGD performance index only evaluate the population's diversity and convergence in the objective space, while the IGDX only evaluate them in the decision space. All experimental results were obtained by running the algorithm 30 times independently on each benchmark problem. IGD can be defined as follows:

$$IGD(O, P^*) = \frac{\sum_{v \in P^*} d(f(v), f(O))}{|P^*|}, \tag{9}$$

where $f(x)$ represents the mapping of x in the objective space.

As seen from Table 9, when DSS is only used in the decision space, CDP-BCD-Dec has better IGDX values on most of the benchmark problems but poor IGD values on all of the problems. When DSS is used only in the objective space, CDP-BCD-Obj has better IGD values on most benchmark problems but poor IGDX values on all of the problems. The gap between CDP-BCD-Dec and CDP-BCD in IGDX is relatively small, and the gap between CDP-BCD-Obj and CDP-BCD in IGD is also relatively small. Therefore, to better balance diversity between the

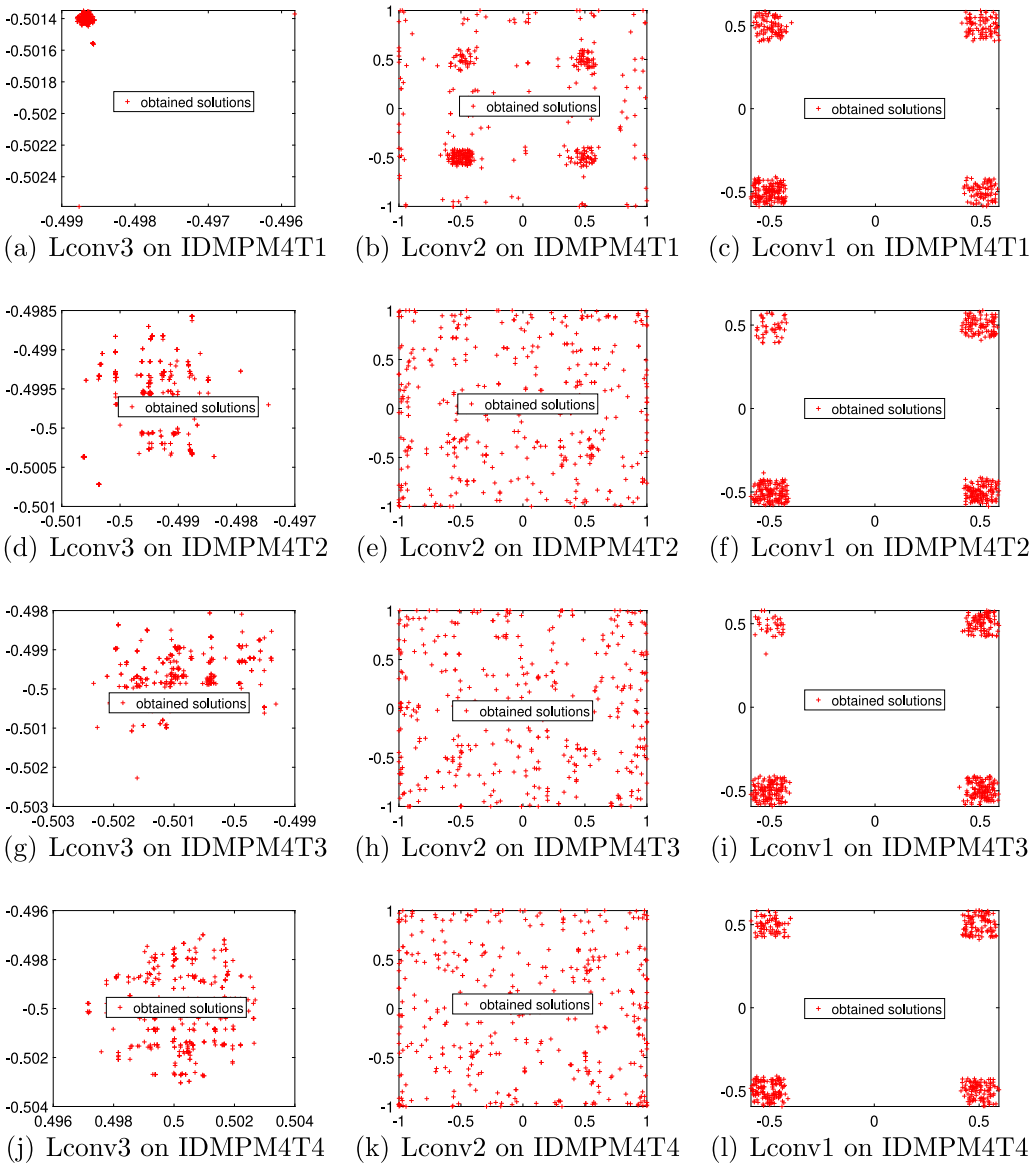


Fig. 14. Ability to locate equivalent PSs in auxiliary populations of different local convergence qualities.

decision and objective spaces, it is necessary to combine the diversity between the decision and objective spaces when using DSS.

5.7. Influence of parameters η and L

The parameter η controls the radius size for calculating strength local convergence quality. The size of η determines whether the algorithm can search for the local PS and imbalanced PS to obtain excellent convergence quality.

Fig. 15 is the average rank of IGDX and IGDM on MMOP-ICDs and Polygon-based MMOPLs obtained by CDP-BCD over 30 runs. It can be seen from Fig. 15 that different η values present different convergence qualities. When η is near 0.2, the algorithm retains the local PS and the imbalanced PS due to the small radius. Therefore, the algorithm has better convergence quality when η is near 0.2. When η is near 1, the algorithm has poor convergence quality because the radius covers the entire decision space so that the algorithm only retains the global PS and the PS with less search difficulty. Therefore, when the decision maker only wants to retain the global PS, η can be set larger. If the decision maker wants to retain the local PS, η can be set smaller.

The L in the NBT component is to control the scope of exploration and development of each solution in the decision space. If L is too large or too small, it will affect the algorithm's performance. Fig. 16 is the average rank of IGDX obtained by CDP-BCD over 30 runs. The benchmark problems selected in Fig. 16 are IDMP, MMF1-8, and Polygon-M6-8 benchmark problems. Each result is obtained from 30 independent runs. It can be seen from Fig. 16 that when L is too large or too small, the performance of CDP-BCD on IGDX will become worse. When L is too small, multiple solutions will develop an area at the same time, resulting in a waste of exploration and development resources. When L is too large, some areas will be missed, resulting in poor diversity of PS or even loss of PS. Therefore, to get better performance, we set the value of L to 3.

5.8. Influence of function evaluations

The comparison algorithms in this section have different settings for the maximum number of function evaluations. The maximum number of function evaluations will affect the final effect of the algorithm. The purpose of this article to set the maximum number of function evaluations to 10000*D is to enable comparisons after the comparison

Table 8
Ablation experiment results of NBT on the MMOP-ICD and Polygon-based MMOPL.

Problem	CDP-BCD_NoNBT	CDP-BCD	CDP-BCD_NoNBT	CDP-BCD
	IGDX		IGDM	
IDMPM2T1	5.9191e−4 (1.74e−5) −	**5.8431e−4 (2.10e−5)**	4.4392e−3 (3.55e−4) −	**4.2262e−3 (2.24e−4)**
IDMPM2T2	8.2751e−4 (6.15e−5) −	**7.4463e−4 (6.63e−5)**	4.2630e−3 (1.36e−4) −	**4.1175e−3 (1.62e−4)**
IDMPM2T3	1.1565e−3 (6.48e−5) =	**1.1556e−3 (1.15e−4)**	**4.6196e−3 (1.21e−4)** +	5.3422e−3 (7.21e−4)
IDMPM2T4	**5.8607e−4 (1.22e−5)** =	5.8689e−4 (2.45e−5)	4.1429e−3 (2.01e−4) =	**4.1152e−3 (2.17e−4)**
IDMPM3T1	7.1356e−3 (1.05e−4) −	**6.6021e−3 (8.10e−5)**	5.2299e−2 (8.90e−4) −	**4.7278e−2 (7.00e−4)**
IDMPM3T2	7.2977e−3 (8.12e−5) −	**6.6477e−3 (7.01e−5)**	5.1772e−2 (6.95e−4) −	**4.6524e−2 (5.09e−4)**
IDMPM3T3	7.9842e−3 (1.32e−4) −	**7.1649e−3 (8.81e−5)**	5.3515e−2 (9.78e−4) −	**4.7479e−2 (6.32e−4)**
IDMPM3T4	7.0598e−3 (9.48e−5) −	**6.5511e−3 (8.67e−5)**	5.0971e−2 (8.68e−4) −	**4.6540e−2 (6.36e−4)**
IDMPM4T1	**7.1612e−3 (8.63e−5)** +	7.4473e−3 (4.62e−4)	6.1877e−2 (3.03e−3) −	**5.8820e−2 (4.32e−3)**
IDMPM4T2	**7.6473e−3 (8.03e−4)** +	8.2391e−3 (1.76e−3)	**5.9990e−2 (9.06e−3)** =	6.2297e−2 (1.37e−2)
IDMPM4T3	8.6248e−3 (1.83e−4) −	**8.3697e−3 (3.45e−4)**	5.7924e−2 (1.51e−3) −	**5.5032e−2 (2.93e−3)**
IDMPM4T4	7.6209e−3 (1.43e−4) −	**7.2141e−3 (1.73e−4)**	5.4964e−2 (1.93e−3) −	**5.0623e−2 (1.78e−3)**
IDMPM2T1_e	5.8115e−4 (1.45e−5) =	**5.7673e−4 (1.24e−5)**	**4.0956e−3 (2.91e−4)** =	4.1050e−3 (2.29e−4)
IDMPM2T2_e	**8.1264e−4 (4.74e−5)** =	8.1481e−4 (5.26e−5)	4.0169e−3 (8.49e−5) =	**4.0148e−3 (1.03e−4)**
IDMPM2T3_e	**8.3690e−4 (2.38e−5)** +	8.5870e−4 (3.80e−5)	**5.9527e−3 (1.34e−4)** +	6.1176e−3 (2.10e−4)
IDMPM2T4_e	**4.2501e−2 (1.70e−5)** +	4.2538e−2 (4.13e−5)	**1.8289e−2 (2.12e−4)** +	1.8816e−2 (4.12e−4)
IDMPM3T1_e	7.2227e−3 (1.12e−4) −	**6.8838e−3 (9.39e−5)**	4.4822e−2 (8.92e−4) −	**4.2083e−2 (6.86e−4)**
IDMPM3T2_e	4.8456e−3 (7.23e−5) −	**4.7795e−3 (8.61e−5)**	3.7985e−2 (6.64e−4) −	**3.7252e−2 (7.77e−4)**
IDMPM3T3_e	5.3222e−3 (8.90e−5) −	**5.2087e−3 (1.66e−4)**	3.9388e−2 (7.09e−4) −	**3.8346e−2 (1.21e−3)**
IDMPM3T4_e	7.2423e−3 (9.89e−5) −	**7.1524e−3 (1.86e−4)**	5.8043e−2 (7.65e−4) −	**5.6516e−2 (1.54e−3)**
Polygon-2	**8.3260e−4 (2.93e−5)** +	8.6125e−4 (4.66e−5)	**2.9435e−3 (1.04e−4)** +	3.0437e−3 (1.65e−4)
Polygon-3	1.7823e−2 (4.45e−4) =	**1.7739e−2 (5.78e−4)**	4.0390e−2 (9.70e−4) =	**4.0261e−2 (1.24e−3)**
Polygon-4	2.0828e−2 (4.51e−4) −	**2.0579e−2 (4.32e−4)**	4.8357e−2 (9.89e−4) =	**4.7921e−2 (1.01e−3)**
Polygon-5	1.9400e−2 (3.07e−4) −	**1.9078e−2 (2.61e−4)**	7.0002e−2 (1.07e−3) −	**6.8573e−2 (9.08e−4)**
Polygon-6	1.9961e−2 (3.24e−4) −	**1.9559e−2 (2.20e−4)**	7.4974e−2 (1.23e−3) −	**7.2928e−2 (8.61e−4)**
Polygon-8	2.0488e−2 (3.62e−4) −	**2.0014e−2 (3.64e−4)**	8.8481e−2 (1.53e−3) −	**8.5742e−2 (1.50e−3)**
Polygon-10	2.0807e−2 (4.24e−4) −	**2.0262e−2 (2.99e−4)**	1.0030e−1 (2.06e−3) −	**9.6747e−2 (1.38e−3)**
Polygon-12	2.0982e−2 (3.72e−4) −	**2.0336e−2 (3.62e−4)**	1.1039e−1 (1.85e−3) −	**1.0629e−1 (1.52e−3)**
+/−/=	5/18/5		4/18/6	

algorithms have achieved the best results, so that the comparison experiments can be as fair as possible. As shown in Fig. 17, when the number of evaluations is 10,000, the performance of all algorithms in IGDX has not yet reached the best results. However, when the number of evaluations is 20,000, the performance of all algorithms in IGDX tends to be stable and reaches the best results. Therefore, to ensure that all algorithms can show the best results when doing comparative experiments, this article sets the maximum number of function evaluations for all problems to 10000*D (D is the dimension of the decision space).

5.9. Application to location selection problem

Fig. 18 is a practical application problem of map-based rental location selection. In this scenario, the tenant wants to find a location close to schools, hospitals, and stores. Therefore, this is a distance minimization problem with three objectives: the distance between the selected location and the school, and the distance between the selected location and the hospital, the distance between the selected location and the store. Three equivalent PSs are in Fig. 18, triangles A, B, and C. Since the sizes of the three triangles are different, there will be some local PSs, such as triangles B and C. Although the solutions in the global PSs (triangle A) are better than those in the local PS regarding distance, the tenant hopes there will be a business district near the rental location. Therefore, according to the tenant's need, $Local\ PSs$ (triangles B and C) will be the most suitable for the tenant, although $Local\ PS$ is an optimal local option.

As seen from Fig. 19, CDP-BCD, RHEA, and HREA can search the locations of most PSs. But only CDP-BCD can cover more Pareto optimal solutions, so it can be seen from Table 10 that CDP-BCD has good advantages in the decision space. Since HREA is only suitable for solving MMOPs where the global PF and local PF are close to each other, most of the solutions of HREA are located on the global PS. MO_Ring_SCD_PSO, TriMOEA-TA&R, CPDEA, and MMEA-WI are designed to solve MMOPs with only global PS, they only search the global PS. However, it can be seen from Fig. 19 that when there is an intersection between two PSs, CDP-BCD's diversity in the decision space becomes worse. The variation in diversity in the decision space is due to the use of SLCQ, which creates an imbalance between diversity and convergence in the intersecting PS parts. In summary, our proposed CDP-BCD is highly competitive in solving MMOPs.

6. Conclusion

The key point in solving MMOPs is balancing convergence and diversity in the decision space. If the convergence and diversity in the decision can be well balanced during evolution, it is very beneficial for the algorithm to solve MMOPs. CDP-BCD balances convergence and diversity in the decision space by relaxing the convergence-first mechanism. The advantages of CDP-BCD are mainly reflected in the new dual-population weak interaction coevolutionary mechanism (DPCE) and the strength local convergence quality (SLCQ). DPCE weakens the interaction intensity between the main population and the auxiliary population, allowing the algorithm to search for PSs that are more difficult to search. SLCQ enhances the selection pressure on the algorithm's

Table 9
Experimental results of DSS in decision space, objective space and dual space.

Problem	CDP-BCD-Obj	CDP-BCD-Dec	CDP-BCD	CDP-BCD-Obj	CDP-BCD-Dec	CDP-BCD
	IGDX			IGD		
IDMPM2T1	9.8113e−4 (3.23e−4) −	5.9847e−4 (6.92e−5) =	**5.8431e−4 (2.10e−5)**	**4.3856e−4 (2.99e−5)** +	5.1595e−4 (2.29e−5) =	5.1515e−4 (2.07e−5)
IDMPM2T2	9.3780e−4 (2.39e−4) −	8.0924e−4 (4.68e−5) −	**7.4463e−4 (6.63e−5)**	**4.0730e−4 (9.40e−6)** +	5.3348e−4 (2.32e−5) −	5.0463e−4 (2.05e−5)
IDMPM2T3	3.2007e−3 (1.10e−3) −	1.2749e−3 (1.18e−4) −	**1.1556e−3 (1.15e−4)**	**4.1342e−4 (7.78e−6)** +	5.5976e−4 (1.63e−5) −	5.1270e−4 (1.97e−5)
IDMPM2T4	1.0243e−3 (5.29e−4) −	6.0183e−4 (8.19e−5) =	**5.8689e−4 (2.45e−5)**	**4.1537e−4 (2.19e−5)** +	5.0546e−4 (2.96e−5) =	4.9698e−4 (1.51e−5)
IDMPM3T1	7.5891e−3 (1.91e−4) −	**6.5937e−3 (7.88e−5)** =	6.6021e−3 (8.10e−5)	**4.1560e−3 (1.21e−4)** +	4.5848e−3 (1.06e−4) =	4.5631e−3 (1.07e−4)
IDMPM3T2	7.6973e−3 (1.86e−4) −	6.6672e−3 (1.04e−4) =	**6.6477e−3 (7.01e−5)**	**3.9694e−3 (6.10e−5)** +	4.4508e−3 (9.23e−5) =	4.4526e−3 (7.36e−5)
IDMPM3T3	8.7438e−3 (3.54e−4) −	7.2580e−3 (8.88e−5) −	**7.1649e−3 (8.81e−5)**	**4.0559e−3 (7.61e−5)** +	4.5675e−3 (7.62e−5) −	4.4883e−3 (6.71e−5)
IDMPM3T4	7.6078e−3 (2.11e−4) −	**6.5311e−3 (6.59e−5)** =	6.5511e−3 (8.67e−5)	**3.9829e−3 (1.10e−4)** +	4.4192e−3 (6.30e−5) =	4.4436e−3 (7.82e−5)
IDMPM4T1	8.5360e−3 (4.69e−4) −	**7.2401e−3 (3.28e−4)** =	7.4473e−3 (4.62e−4)	**5.9629e−3 (3.86e−4)** +	6.5082e−3 (3.13e−4) =	6.4475e−3 (3.50e−4)
IDMPM4T2	1.7669e−2 (4.85e−2) −	**7.6332e−3 (4.44e−4)** =	8.2391e−3 (1.76e−3)	**5.4359e−3 (2.21e−4)** +	5.8683e−3 (2.03e−4) −	5.7764e−3 (1.97e−4)
IDMPM4T3	1.8549e−2 (4.86e−2) −	8.5029e−3 (3.40e−4) =	**8.3697e−3 (3.45e−4)**	**5.4392e−3 (1.51e−4)** +	5.9999e−3 (1.56e−4) −	5.8752e−3 (1.23e−4)
IDMPM4T4	8.1646e−3 (2.22e−4) −	7.2715e−3 (1.91e−4) =	**7.2141e−3 (1.73e−4)**	**5.3579e−3 (2.85e−4)** +	5.8188e−3 (2.05e−4) =	5.7943e−3 (2.11e−4)
MMF1	4.5377e−2 (1.65e−3) −	3.3171e−2 (1.16e−3) −	**3.2372e−2 (8.37e−4)**	**2.2260e−3 (4.56e−5)** +	3.6504e−3 (2.32e−4) −	3.2445e−3 (1.59e−4)
MMF1_e	2.3481e−1 (4.98e−2) =	**2.2056e−1 (5.53e−2)** =	2.2627e−1 (5.94e−2)	**3.1569e−3 (3.46e−4)** +	3.5045e−3 (3.80e−4) =	3.4148e−3 (3.08e−4)
MMF1_z	3.3056e−2 (2.01e−3) −	2.3363e−2 (8.10e−4) −	**2.2717e−2 (5.84e−4)**	**2.2222e−3 (1.01e−4)** +	3.1709e−3 (1.68e−4) −	2.8548e−3 (1.24e−4)
MMF2	2.2693e−2 (8.94e−3) =	**2.2575e−2 (1.58e−2)** =	2.5330e−2 (1.50e−2)	**8.6708e−3 (3.91e−3)** =	9.5269e−3 (6.59e−3) =	1.0892e−2 (1.03e−2)
MMF3	1.7040e−2 (6.23e−3) =	1.9167e−2 (1.11e−2) =	**1.5809e−2 (5.00e−3)**	8.0976e−3 (5.83e−3) =	9.0167e−3 (1.03e−2) =	**6.6532e−3 (2.77e−3)**
MMF4	2.7544e−2 (2.45e−3) −	**1.8863e−2 (5.36e−4)** +	1.9385e−2 (6.02e−4)	**2.5276e−3 (7.15e−5)** +	3.4762e−3 (1.68e−4) −	3.3017e−3 (1.20e−4)
MMF5	7.6969e−2 (3.84e−3) −	5.9367e−2 (1.42e−3) −	**5.8427e−2 (1.66e−3)**	**2.3894e−3 (7.88e−5)** +	3.9975e−3 (2.11e−4) −	3.6523e−3 (2.12e−4)
MMF6	6.7023e−2 (2.55e−3) −	**5.3932e−2 (1.05e−3)** =	5.4070e−2 (1.13e−3)	**2.4004e−3 (5.73e−5)** +	3.7733e−3 (2.11e−4) −	3.6510e−3 (2.04e−4)
MMF7	2.4302e−2 (1.90e−3) −	**1.8884e−2 (7.45e−4)** =	1.9016e−2 (6.59e−4)	**2.2813e−3 (6.71e−5)** +	3.5449e−3 (2.05e−4) −	3.1000e−3 (1.22e−4)
MMF8	5.5754e−2 (5.11e−3) −	3.8071e−2 (2.18e−3) =	**3.7898e−2 (2.44e−3)**	**2.5617e−3 (8.78e−5)** +	3.5430e−3 (1.92e−4) =	3.5685e−3 (2.00e−4)
MMF12	3.4034e−3 (1.77e−4) −	**2.7437e−3 (9.51e−5)** +	3.2590e−3 (1.47e−4)	**8.4153e−2 (2.54e−4)** =	8.5155e−2 (4.92e−4) −	8.4239e−2 (2.27e−4)
MMF14	5.0255e−2 (1.25e−3) −	**3.8866e−2 (5.07e−4)** +	4.0278e−2 (4.97e−4)	**5.9997e−2 (1.40e−3)** +	6.6134e−2 (1.05e−3) −	6.3441e−2 (8.96e−4)
MMF14_a	6.4738e−2 (1.97e−3) −	**4.8727e−2 (4.21e−4)** +	5.2056e−2 (5.53e−4)	**6.2883e−2 (1.93e−3)** +	6.8296e−2 (1.32e−3) −	6.6649e−2 (8.57e−4)
MMF15	4.9193e−2 (7.97e−4) −	**4.3526e−2 (6.63e−4)** +	4.4069e−2 (7.79e−4)	**1.8869e−1 (4.99e−3)** =	1.9462e−1 (4.70e−3) −	1.8916e−1 (4.15e−3)
MMF15_a	6.2566e−2 (1.92e−3) −	**5.2764e−2 (8.95e−4)** +	5.5998e−2 (1.09e−3)	1.8823e−1 (5.23e−3) =	1.8698e−1 (3.28e−3) =	**1.8649e−1 (3.22e−3)**
+/−/=	0/24/3	6/6/15		22/0/5	0/15/12	

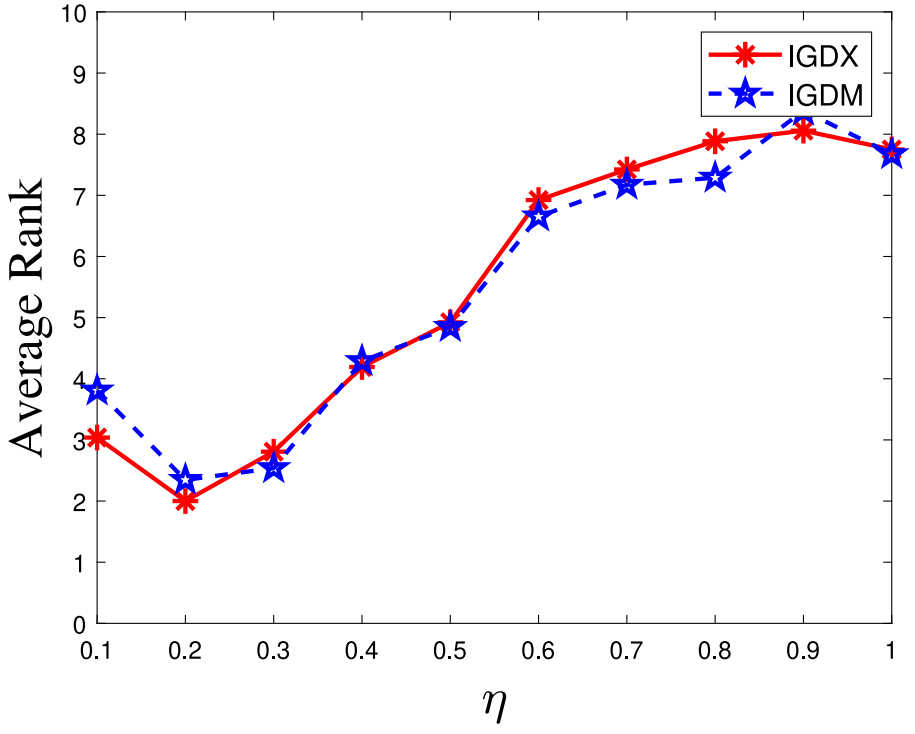


Fig. 15. The average rank of IGDX and IGDM on MMOP-ICDs and Polygon-based MMOPLs obtained by CDP-BCD with different values of η.

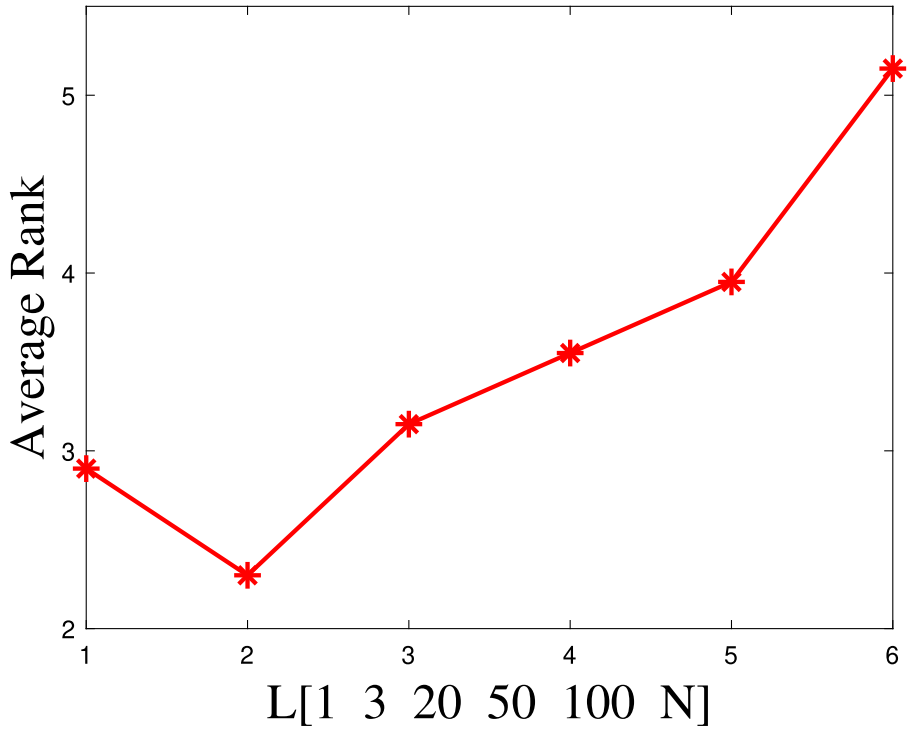


Fig. 16. The average rank of IGDX obtained by CDP-BCD with different values of L.

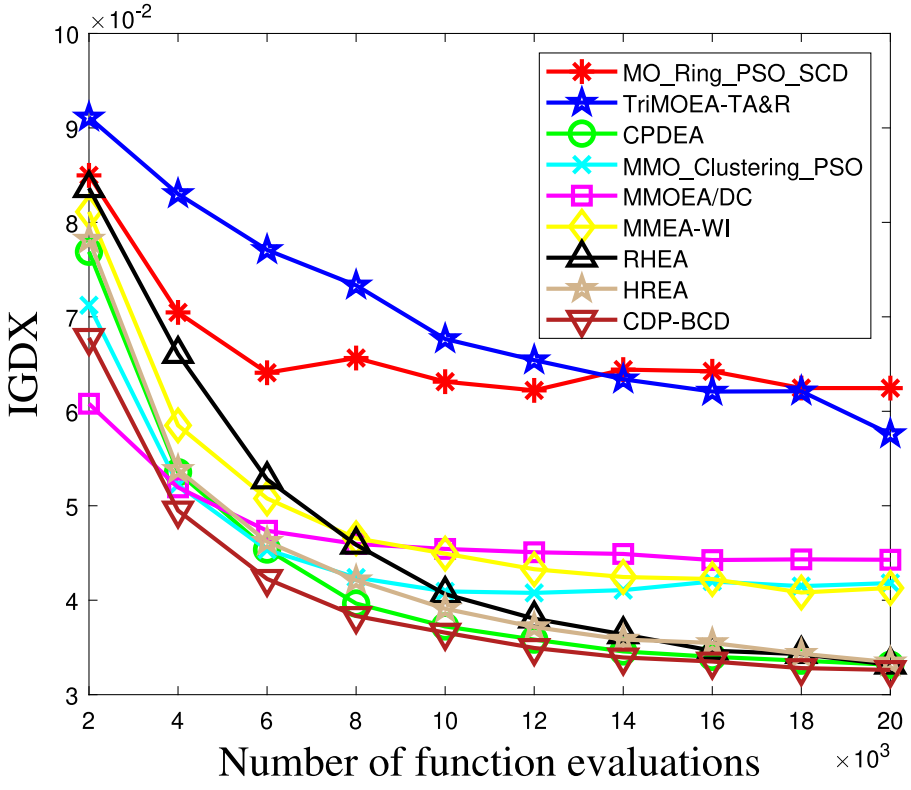


Fig. 17. The IGDX results on MMF1 obtained by comparative algorithms with different numbers of function evaluations.

convergence and improves the population's convergence, especially in high-dimensional space. But CDP-BCD also has its limitations. When the scope of the objective space is more significant than that of the decision space, the ability of CDP-BCD to balance the diversity between the decision space and the objective space will be weakened, as shown in Tables 3 and 5 the results. In addition, if there is an intersection between equivalent PSs, using SLCQ will lead to an imbalance between convergence and diversity in the decision space. One of the two intersecting PSs may have poor diversity. As shown in Fig. 19, the obtained solution does not completely cover part of the true PS.

CDP-BCD performs well on most MMOPs but performs slightly worse on MMOPs with complex manifold PSs (such as the MMF).

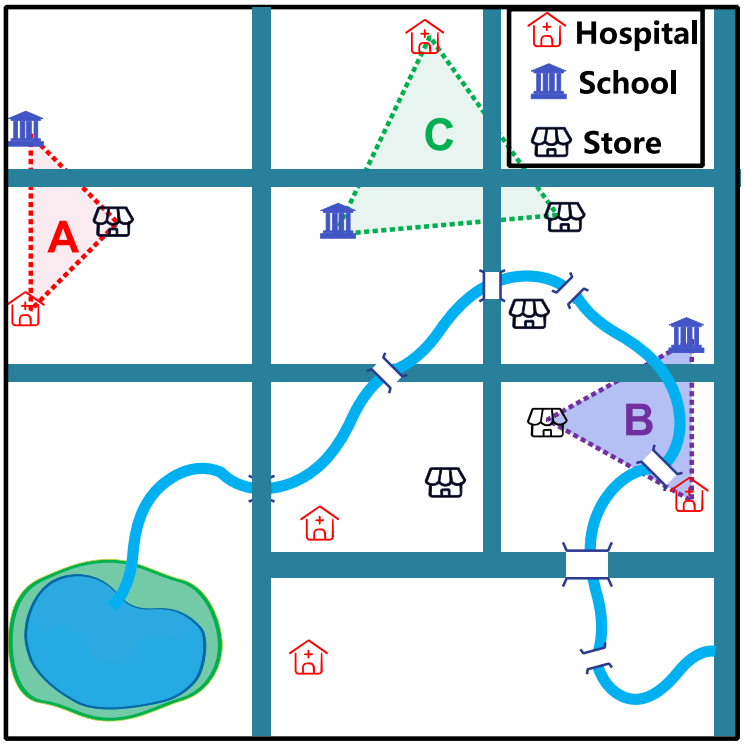


Fig. 18. Map-based rental location selection problem.

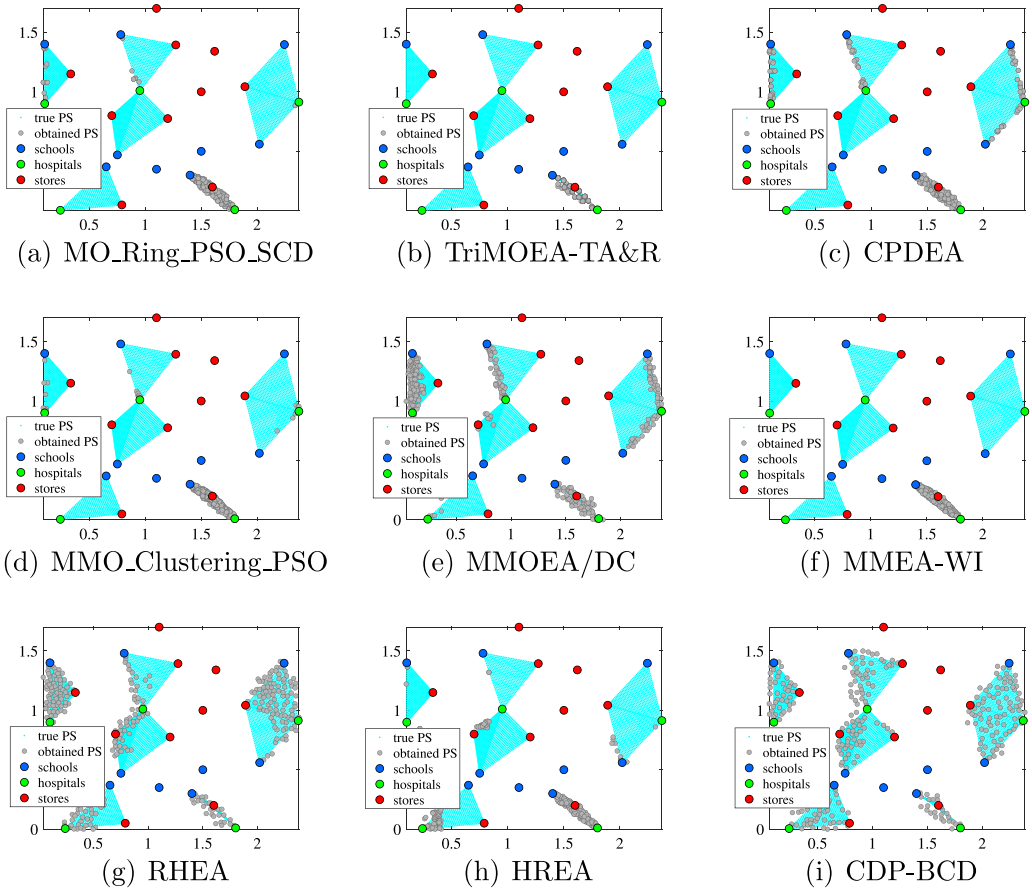


Fig. 19. Distribution of the equivalent PSs obtained by different algorithms on the map-based rental location selection problem.

Table 10
The IGDX and IGDM results obtained by compared algorithms on the map-based rental location selection problem.

Algorithm	MO_Ring_PSO_SCD	TriMOEA-TA&R	CPDEA	MMO_Clustering_PSO	MMOEA/DC
IGDX	1.8363e−1 (3.24e−2) −	7.5831e−1 (1.42e−1) −	1.4273e−1 (2.09e−2) −	2.1752e−1 (3.96e−2) −	9.4874e−2 (3.07e−2) −
IGDM	1.4479e−1 (9.99e−3) −	2.4435e−1 (1.32e−2) −	1.2051e−1 (4.92e−3) −	1.7168e−1 (1.90e−2) −	6.9423e−2 (1.35e−2) −
Algorithm	MMEA-WI	RHEA	HREA	CDP-BCD	+/−/=
IGDX	2.3501e−1 (4.27e−2) −	4.7944e−2 (3.86e−3) −	1.1156e−1 (1.67e−2) −	**3.3632e−2 (7.23e−4)**	0/8/0
IGDM	1.3167e−1 (8.51e−3) −	4.5873e−2 (1.17e−3) =	1.0327e−1 (1.88e−2) −	**4.5336e−2 (5.87e−4)**	0/7/1

Although DSS can cover all PSs, due to the lack of consideration of the distribution contribution of solutions within the niche, DSS needs improvement on MMFs. Most MMOPs are static benchmark problems, and the dimensionality of the objective space is relatively low. There are many dynamic multi-objective optimization problems [50,51] and many-objective optimization problems [52,53] in real life, and future work will focus on the dynamic multimodal multi-objective optimization environment. Finally, combining CDP-BCD with practical application scenarios should lead us to an excellent solution for practical engineering application problems.

CRediT authorship contribution statement

Zhipan Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huigui Rong:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Shengxiang Yang:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization. **Xu Yang:** Writing – review & editing, Methodology. **Yupeng Huang:** Writing – review & editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the Natural Science Foundation of Hunan under Grant 2020JJ4008 and in part by the Major Research Projects of the National Natural Science Foundation of China under Grant 2020GK2006.

References

[1] Y. Han, D. Gong, Y. Jin, Q. Pan, Evolutionary multiobjective blocking lot-streaming flow shop scheduling with machine breakdowns, IEEE Trans. Cybern. 49 (1) (2017) 184–197.
[2] E.B. Tirkolaee, A. Goli, A. Faridnia, M. Soltani, G.-W. Weber, Multi-objective optimization for the reliable pollution-routing problem with cross-dock selection using Pareto-based algorithms, J. Clean. Prod. 276 (2020) 122927.
[3] A. Jaszkiewicz, On the performance of multiple-objective genetic local search on the 0/1 knapsack problem-a comparative experiment, IEEE Trans. Evol. Comput. 6 (4) (2002) 402–412.
[4] K. Deb, Multi-Objective Optimization Using Evolutionary Algorithms, vol. 16, John Wiley & Sons, 2001.
[5] K. Deb, Multi-objective genetic algorithms: Problem difficulties and construction of test problems, Evol. Comput. 7 (3) (1999) 205–230.
[6] C. Yue, B. Qu, K. Yu, J. Liang, X. Li, A novel scalable test problem suite for multimodal multiobjective optimization, Swarm Evol. Comput. 48 (2019) 62–71.
[7] Y. Liu, H. Ishibuchi, G.G. Yen, Y. Nojima, N. Masuyama, Handling imbalance between convergence and diversity in the decision space in evolutionary multimodal multi-objective optimization, IEEE Trans. Evol. Comput. 24 (3) (2019) 551–565.
[8] W. Li, X. Yao, T. Zhang, R. Wang, L. Wang, Hierarchy ranking method for multimodal multi-objective optimization with local Pareto fronts, IEEE Trans. Evol. Comput. (2022).
[9] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, IEEE Trans. Evol. Comput. 6 (2) (2002) 182–197.
[10] E. Zitzler, M. Laumanns, L. Thiele, SPEA2: Improving the strength Pareto evolutionary algorithm, TIK-Rep. 103 (2001).
[11] Q. Zhang, H. Li, MOEA/D: a multiobjective evolutionary algorithm based on decomposition, IEEE Trans. Evol. Comput. 11 (6) (2007) 712–731.
[12] J. Liang, C. Yue, B.-Y. Qu, Multimodal multi-objective optimization: A preliminary study, in: 2016 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2016, pp. 2454–2461.
[13] Y. Liu, H. Ishibuchi, Y. Nojima, N. Masuyama, K. Shang, A double-niched evolutionary algorithm and its behavior on polygon-based problems, in: International Conference on Parallel Problem Solving from Nature, Springer, 2018, pp. 262–273.
[14] C. Yue, B. Qu, J. Liang, A multiobjective particle swarm optimizer using ring topology for solving multimodal multiobjective problems, IEEE Trans. Evol. Comput. 22 (5) (2017) 805–817.
[15] Y. Liu, G.G. Yen, D. Gong, A multimodal multiobjective evolutionary algorithm using two-archive and recombination strategies, IEEE Trans. Evol. Comput. 23 (4) (2018) 660–674.
[16] Z. Li, J. Zou, S. Yang, J. Zheng, A two-archive algorithm with decomposition and fitness allocation for multi-modal multi-objective optimization, Inform. Sci. 574 (2021) 413–430.
[17] W. Li, T. Zhang, R. Wang, H. Ishibuchi, Weighted indicator-based evolutionary algorithm for multimodal multiobjective optimization, IEEE Trans. Evol. Comput. 25 (6) (2021) 1064–1078.
[18] Q. Lin, W. Lin, Z. Zhu, M. Gong, J. Li, C.A.C. Coello, Multimodal multiobjective evolutionary optimization with dual clustering in decision and objective spaces, IEEE Trans. Evol. Comput. 25 (1) (2020) 130–144.
[19] J. Liang, K. Qiao, C. Yue, K. Yu, B. Qu, R. Xu, Z. Li, Y. Hu, A clustering-based differential evolution algorithm for solving multimodal multi-objective optimization problems, Swarm Evol. Comput. 60 (2021) 100788.
[20] G. Li, W. Wang, W. Zhang, Z. Wang, H. Tu, W. You, Grid search based multi-population particle swarm optimization algorithm for multimodal multi-objective optimization, Swarm Evol. Comput. 62 (2021) 100843.
[21] C. Yue, P.N. Suganthan, J. Liang, B. Qu, K. Yu, Y. Zhu, L. Yan, Differential evolution using improved crowding distance for multimodal multiobjective optimization, Swarm Evol. Comput. 62 (2021) 100849.
[22] J. Zou, Q. Deng, Y. Liu, X. Yang, S. Yang, J. Zheng, A dynamic-niching-based Pareto domination for multimodal multiobjective optimization, IEEE Trans. Evol. Comput. (2023).
[23] F. Ming, W. Gong, L. Wang, L. Gao, Balancing convergence and diversity in objective and decision spaces for multimodal multi-objective optimization, IEEE Trans. Emerg. Top. Comput. Intell. 7 (2) (2022) 474–486.
[24] J. Zhang, J. Zou, S. Yang, J. Zheng, An evolutionary algorithm based on independently evolving sub-problems for multimodal multi-objective optimization, Inform. Sci. 619 (2023) 908–929.
[25] M. Xiong, W. Xiong, Z. Liu, Y. Liu, C. Han, A multi-modal multi-objective evolutionary algorithm based on dual decomposition and subset selection, Swarm Evol. Comput. 84 (2024) 101431.
[26] G. Li, M. Sun, Y. Wang, W. Wang, W. Zhang, C. Yue, G. Zhang, A ring-hierarchy-based evolutionary algorithm for multimodal multi-objective optimization, Swarm Evol. Comput. 81 (2023) 101352.
[27] W. Wang, G. Li, Y. Wang, F. Wu, W. Zhang, L. Li, Clearing-based multimodal multi-objective evolutionary optimization with layer-to-layer strategy, Swarm Evol. Comput. 68 (2022) 100976.

[28] G. Li, W. Wang, C. Yue, W. Zhang, Y. Wang, Two-stage adaptive differential evolution with dynamic dual-populations for multimodal multi-objective optimization with local Pareto solutions, Inform. Sci. (2023) 119271.
[29] C. Yang, T. Wu, J. Ji, Two-stage species conservation for multimodal multi-objective optimization with local Pareto sets, Inform. Sci. 639 (2023) 118990.
[30] Z. Lv, S. Li, H. Sun, H. Zhang, A multimodal multi-objective evolutionary algorithm with two-stage dual-indicator selection strategy, Swarm Evol. Comput. (2023) 101319.
[31] B. Qu, G. Li, L. Yan, J. Liang, C. Yue, K. Yu, O.D. Crisalle, A grid-guided particle swarm optimizer for multimodal multi-objective problems, Appl. Soft Comput. 117 (2022) 108381.
[32] Y. Tian, T. Zhang, J. Xiao, X. Zhang, Y. Jin, A coevolutionary framework for constrained multiobjective optimization problems, IEEE Trans. Evol. Comput. 25 (1) (2021) 102–116, http://dx.doi.org/10.1109/TEVC.2020.3004012.
[33] K. Li, R. Chen, G. Fu, X. Yao, Two-archive evolutionary algorithm for constrained multiobjective optimization, IEEE Trans. Evol. Comput. 23 (2) (2019) 303–315, http://dx.doi.org/10.1109/TEVC.2018.2855411.
[34] W. Huang, J. Zou, H. Tang, J. Zheng, F. Yu, Enhanced auxiliary population search for diversity improvement of constrained multiobjective coevolutionary optimization, Swarm Evol. Comput. 83 (2023) 101404.
[35] H.K. Singh, K.S. Bhattacharjee, T. Ray, Distance-based subset selection for benchmarking in evolutionary multi/many-objective optimization, IEEE Trans. Evol. Comput. 23 (5) (2018) 904–912.
[36] N. Beume, B. Naujoks, M. Emmerich, SMS-EMOA: Multiobjective selection based on dominated hypervolume, European J. Oper. Res. 181 (3) (2007) 1653–1669.
[37] W. Chen, H. Ishibuchi, K. Shang, Lazy greedy hypervolume subset selection from large candidate solution sets, in: 2020 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2020, pp. 1–8.
[38] R. Cheng, Y. Jin, M. Olhofer, B. Sendhoff, A reference vector guided evolutionary algorithm for many-objective optimization, IEEE Trans. Evol. Comput. 20 (5) (2016) 773–791.
[39] Y. Liu, H. Ishibuchi, Y. Nojima, N. Masuyama, Y. Han, Searching for local Pareto optimal solutions: A case study on polygon-based problems, in: 2019 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2019, pp. 896–903.
[40] M. Köppen, K. Yoshida, Substitute distance assignments in NSGA-II for handling many-objective optimization problems, in: International Conference on Evolutionary Multi-Criterion Optimization, Springer, 2007, pp. 727–741.
[41] H. Ishibuchi, Y. Hitotsuyanagi, N. Tsukamoto, Y. Nojima, Many-objective test problems to visually examine the behavior of multiobjective evolution in a decision space, in: International Conference on Parallel Problem Solving from Nature, Springer, 2010, pp. 91–100.
[42] W. Wang, G. Li, MMO-ClusteringPSO: competition on multimodal multi-objective optimization, in: 2019 IEEE Congress on Evolutionary Computation, CEC, IEEE, 2021.
[43] K. Deb, R.B. Agrawal, et al., Simulated binary crossover for continuous search space, Complex Syst. 9 (2) (1995) 115–148.
[44] K. Deb, M. Goyal, A combined genetic adaptive search (GeneAS) for engineering design, Comput. Sci. Inform. 26 (1996) 30–45.
[45] D.A. Van Veldhuizen, G.B. Lamont, On measuring multiobjective evolutionary algorithm performance, in: Proceedings of the 2000 Congress on Evolutionary Computation. CEC00 (Cat. No. 00TH8512), vol. 1, IEEE, 2000, pp. 204–211.
[46] P.A. Bosman, D. Thierens, The balance between proximity and diversity in multiobjective evolutionary algorithms, IEEE Trans. Evol. Comput. 7 (2) (2003) 174–188.
[47] J. Bader, E. Zitzler, Hype: An algorithm for fast hypervolume-based many-objective optimization, Evolut. Comput. 19 (1) (2011) 45–76.
[48] A. Zhou, Q. Zhang, Y. Jin, Approximating the set of Pareto-optimal solutions in both the decision and objective spaces by an estimation of distribution algorithm, IEEE Trans. Evol. Comput. 13 (5) (2009) 1167–1189.
[49] J. Derrac, S. García, D. Molina, F. Herrera, A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms, Swarm Evol. Comput. 1 (1) (2011) 3–18.
[50] A. Zhou, Y. Jin, Q. Zhang, A population prediction strategy for evolutionary dynamic multiobjective optimization, IEEE Trans. Cybern. 44 (1) (2013) 40–53.
[51] R. Chen, K. Li, X. Yao, Dynamic multiobjectives optimization with a changing number of objectives, IEEE Trans. Evol. Comput. 22 (1) (2017) 157–171.
[52] K. Li, K. Deb, Q. Zhang, S. Kwong, An evolutionary many-objective optimization algorithm based on dominance and decomposition, IEEE Trans. Evol. Comput. 19 (5) (2014) 694–716.
[53] K. Deb, H. Jain, An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part I: solving problems with box constraints, IEEE Trans. Evol. Comput. 18 (4) (2013) 577–601.